Handheld radio deveploment Board

Rev 0.1

Zettaone Technologies
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Title:	Handheld Radio Deveploment Board
Sheet Name:	Macnica_Handheld Radio Deveploment Board
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 1 of 31
Author: Arun P	Reviewed by: <Reviewed By>

D C B A

Revision History - Rev 0.1

Sl. No	Version	Description	Date
1	V0.1	Initial Draft	03-07-2027

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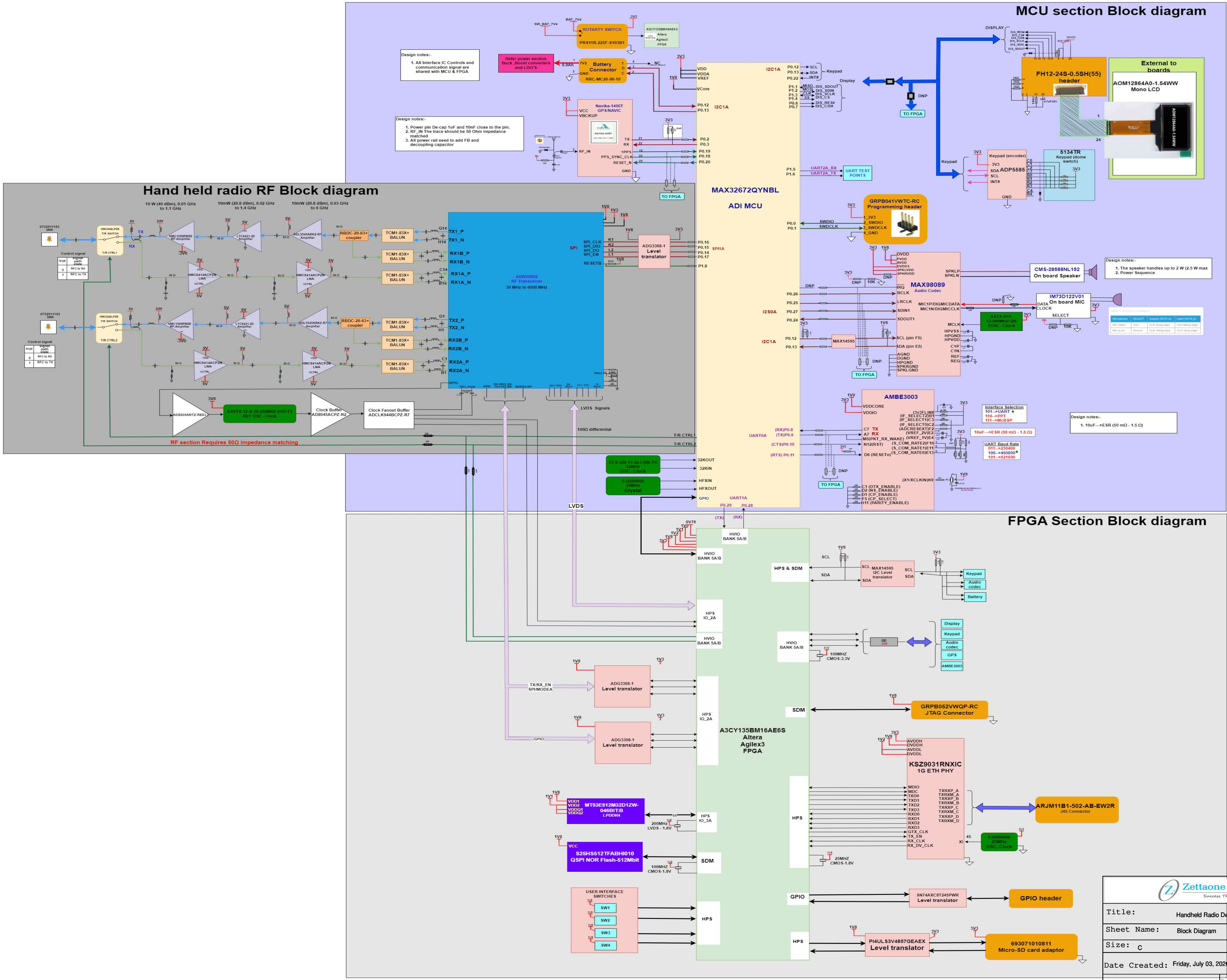
Title: Handheld Radio Depeloment Board	
Sheet Name: Revision History	
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 2 of 31
Author: Arun P	Reviewed by: <Reviewed By>

TABLE OF CONTENT

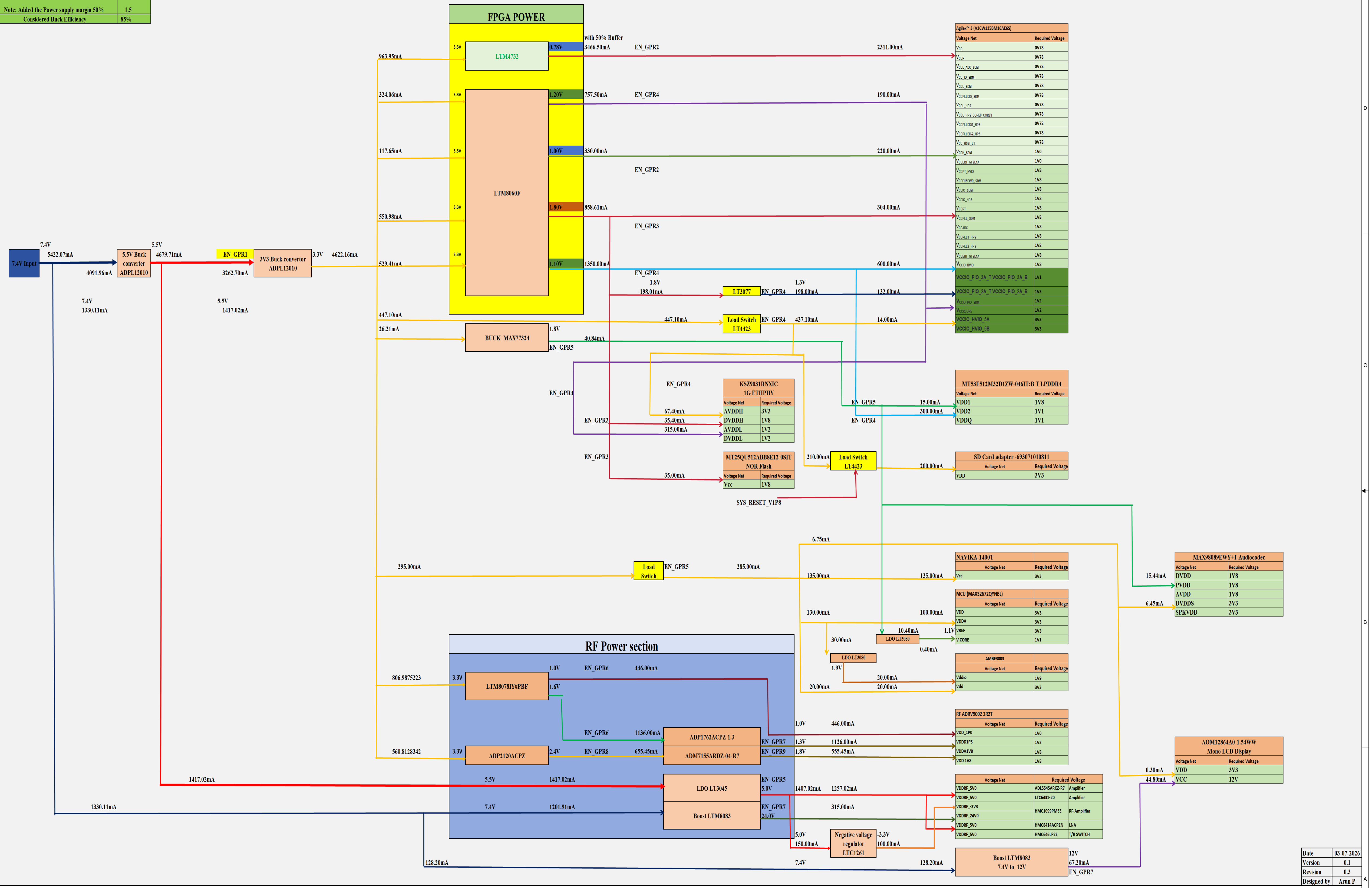
Hand held radio architecture Block diagram



Zettaone Technologies — Success Through Agile Technologies...	
Title:	Handheld Radio Development Board
Sheet Name:	Block Diagram
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 4 of 31
Author: Arun P	Reviewed by: <Reviewed By>

Handheld Radio Board Power Scheme

Note: Added the Power supply margin 50%	1.5
Considered Buck Efficiency	85%



Date	03-07-2026
Version	0.1
Revision	0.3
Designed by	Arun P

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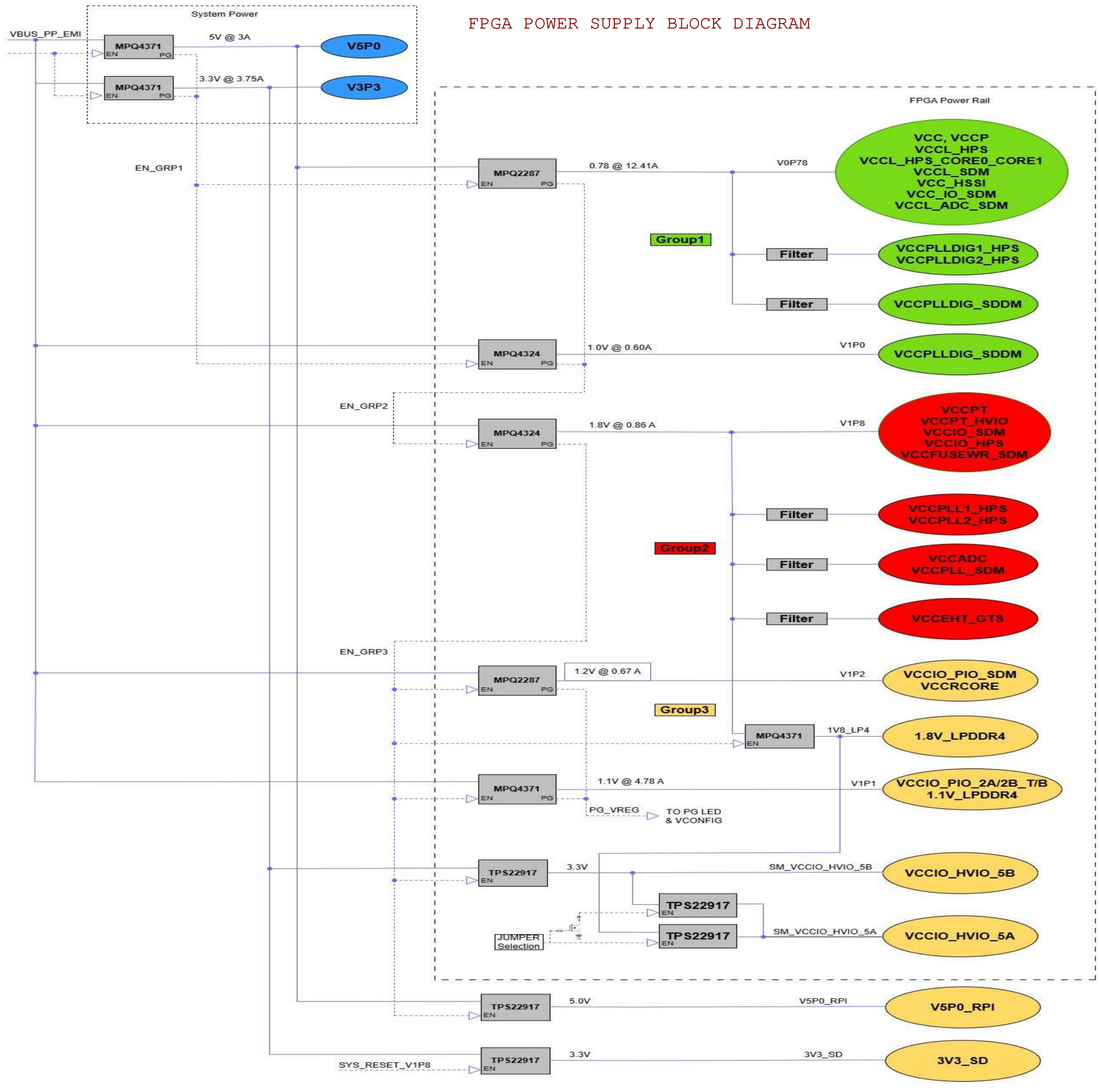
Title:	Handheld Radio Development Board
Sheet Name:	Clock Tree
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 5 of 31
Author: Arun P	Reviewed by: <Reviewed By>

FPGA POWER SUPPLY BLOCK DIAGRAM



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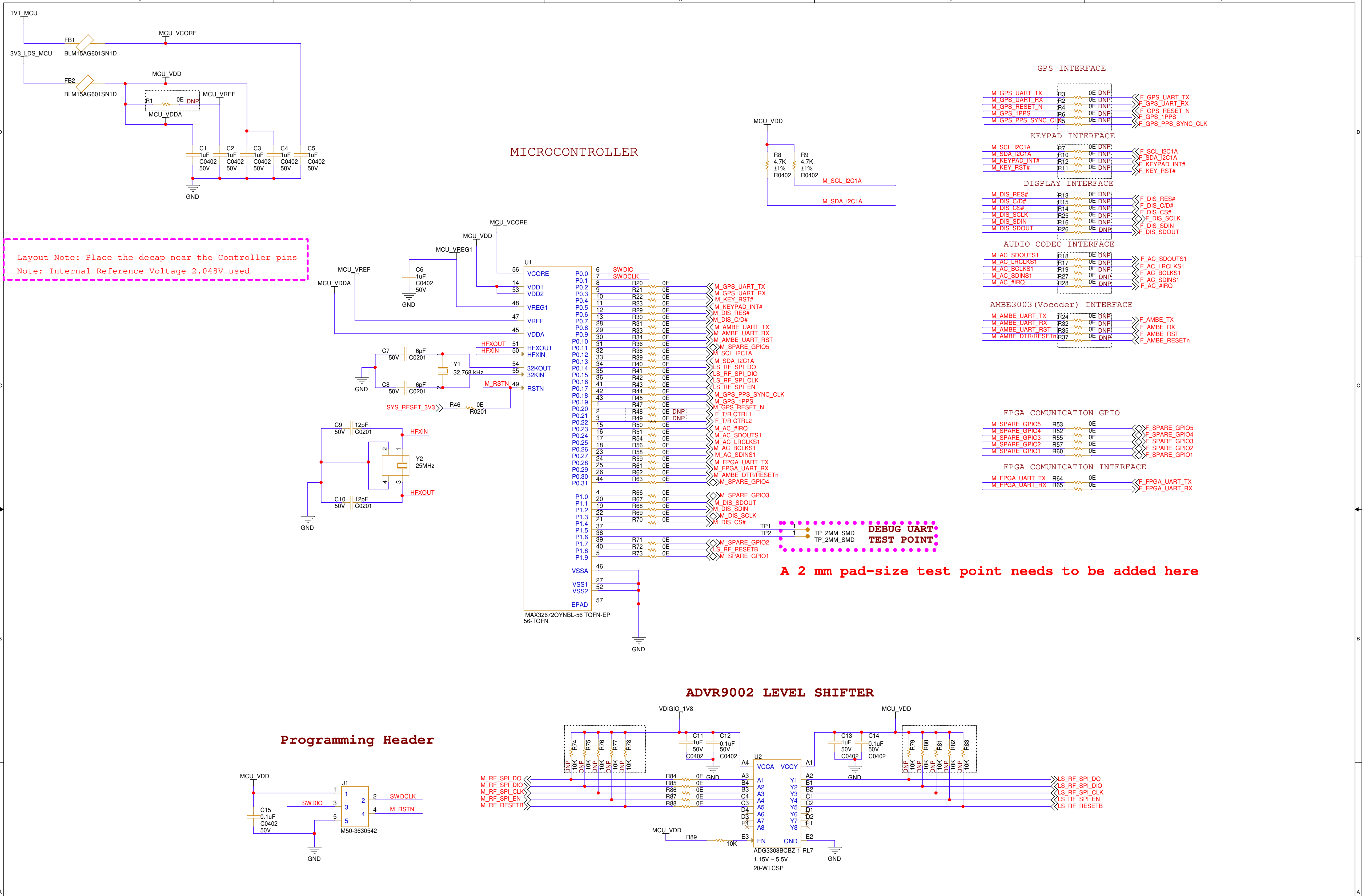
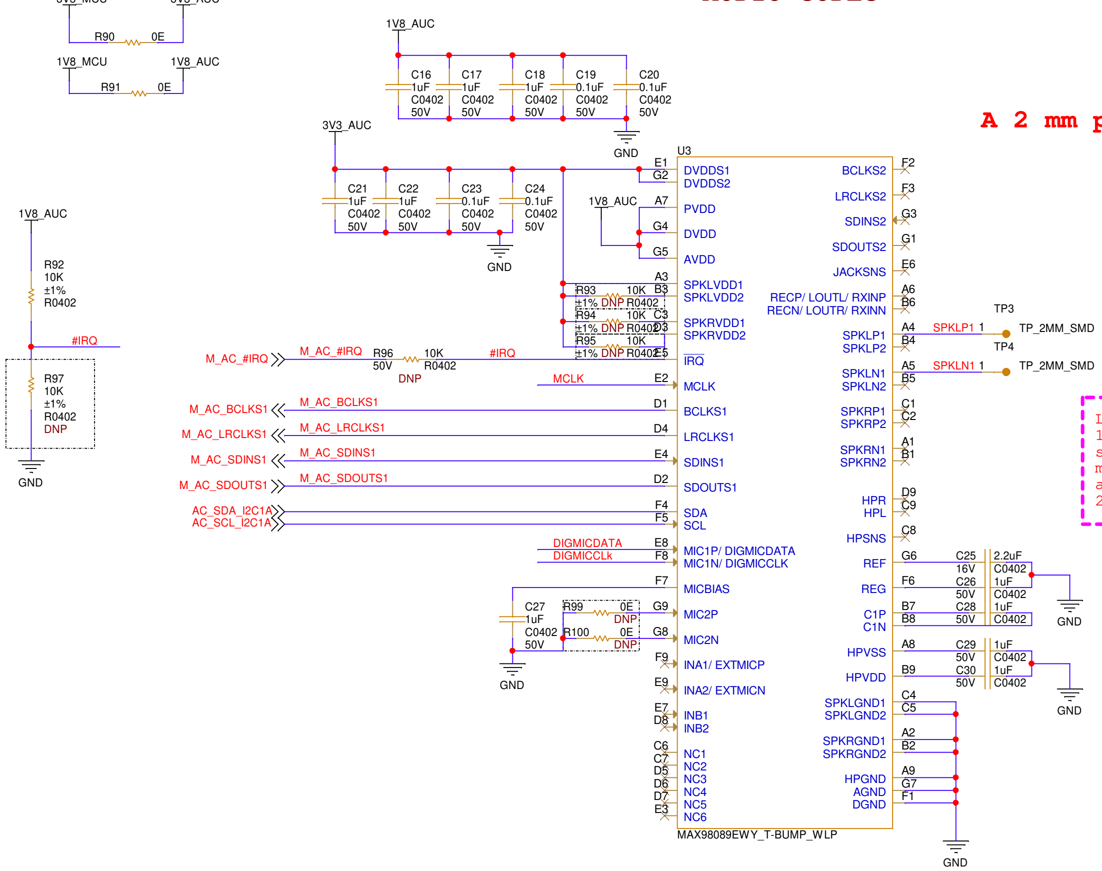
Title:	Handheld Radio Development Board
Sheet Name:	FPGA Power Supply Block Diagram
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 6 of 31
Author: Arun P	Reviewed by: <Reviewed By>



Zettaone Technologies — Success Through Agile Technologies...			
Title:	Handheld Radio Deveplopment Board		
Sheet Name:	Microcontroller		
Size:	C	REV:	R0.1
Date Created:	Friday, July 03, 2026	Sheet:	7 of 31
Author:	Arun P	Reviewed by:	<Reviewed By>

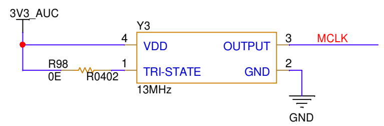
AUDIO CODEC



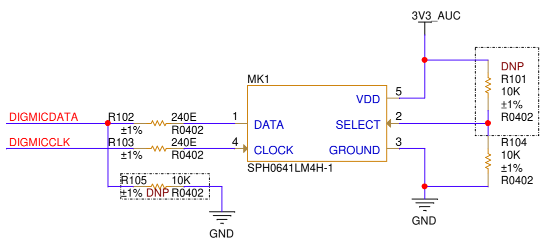
A 2 mm pad-size test point needs to be added here

SPEAKER

Layout Note:
1. Provide solder pads for the speaker wire connection (TP) and maintain a minimum 3 mm handling clearance around the speaker TP.
2.Route SPKLP/SPKLN as a differential pair



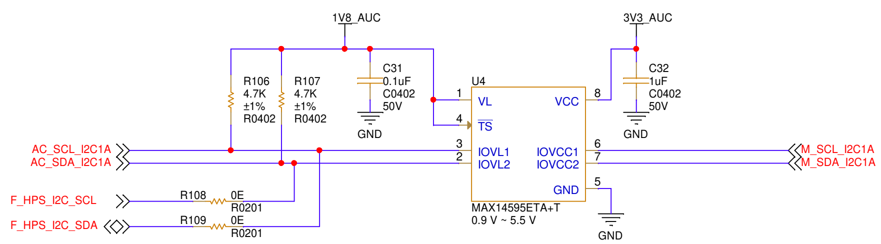
MICROPHONE



I2C Address

Address Format	Value
7-bit I²C address	0x10
8-bit Write	0x20
8-bit Read	0x21

I2C LEVEL TRANSLATOR



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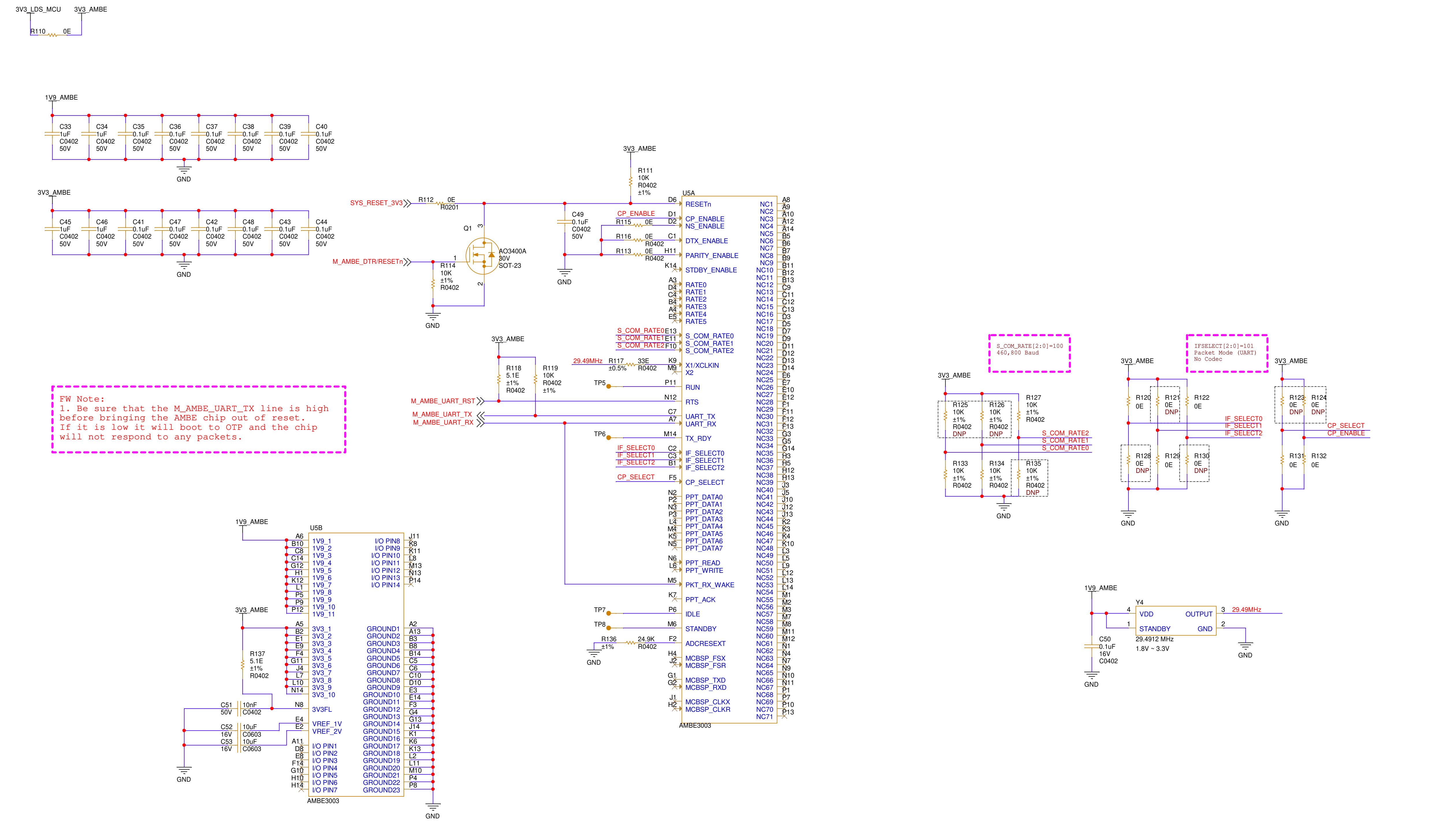
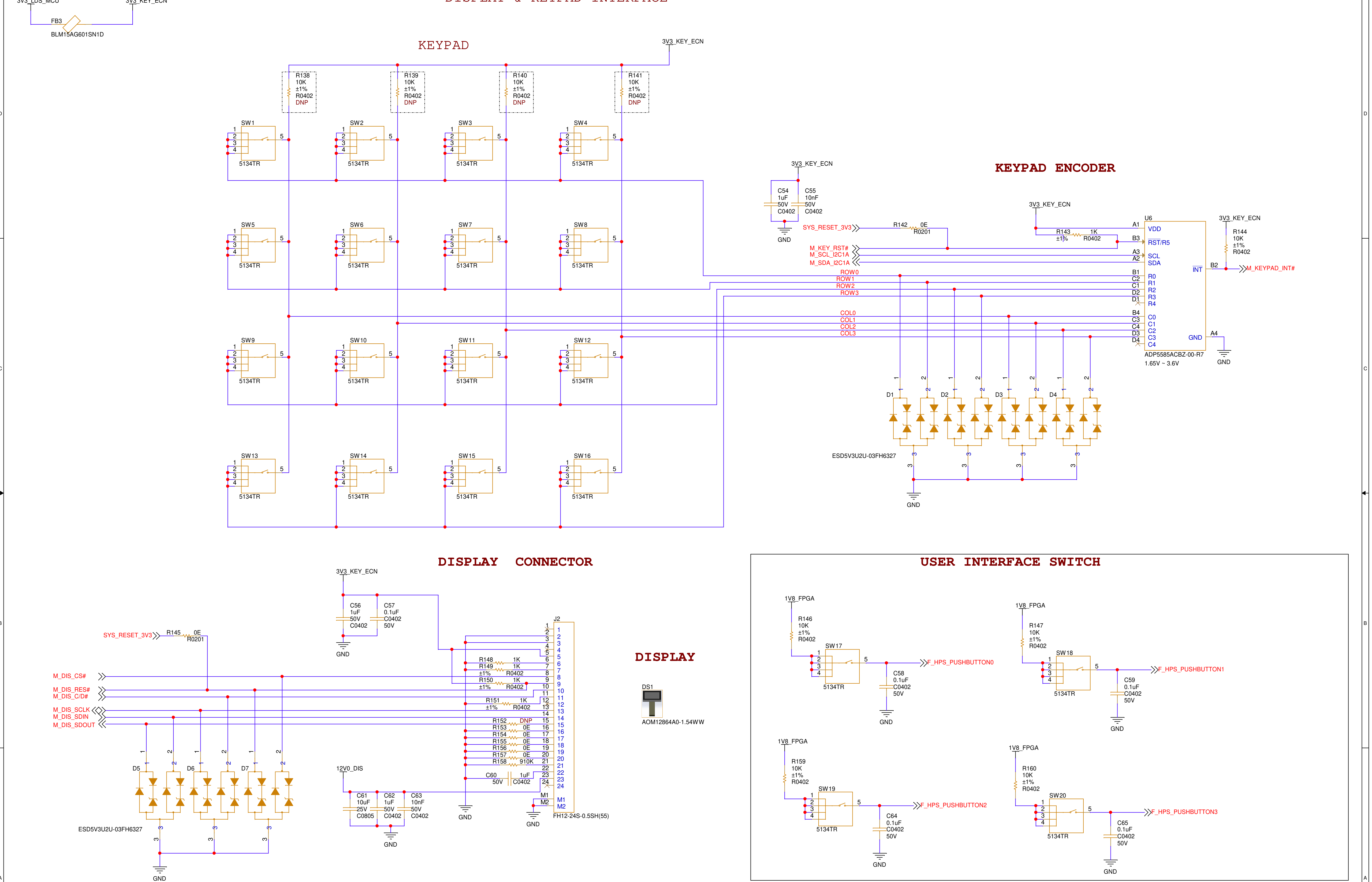
Title:	Handheld Radio Development Board		
Sheet Name:	Aduiocodec,Microphone & Speaker Interface		
Size:	C	REV:	R0.1
Date Created:	Friday, July 03, 2026	Sheet:	8 of 31
Author:	Arun P	Reviewed by:	<Reviewed By>

AMBE3003 - VOCODER INTERFACE



Zettaone Technologies — Success Through Agile Technologies...	
Title:	Handheld Radio Development Board
Sheet Name:	AMBE3003 - VOCODER Interface
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 9 of 31
Author: Arun P	Reviewed by: <Reviewed By>

DISPLAY & KEYPAD INTERFACE



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Title:	Handheld Radio Development Board		
Sheet Name:	Display & Keypad Interface		
Size:	C	REV:	R0.1
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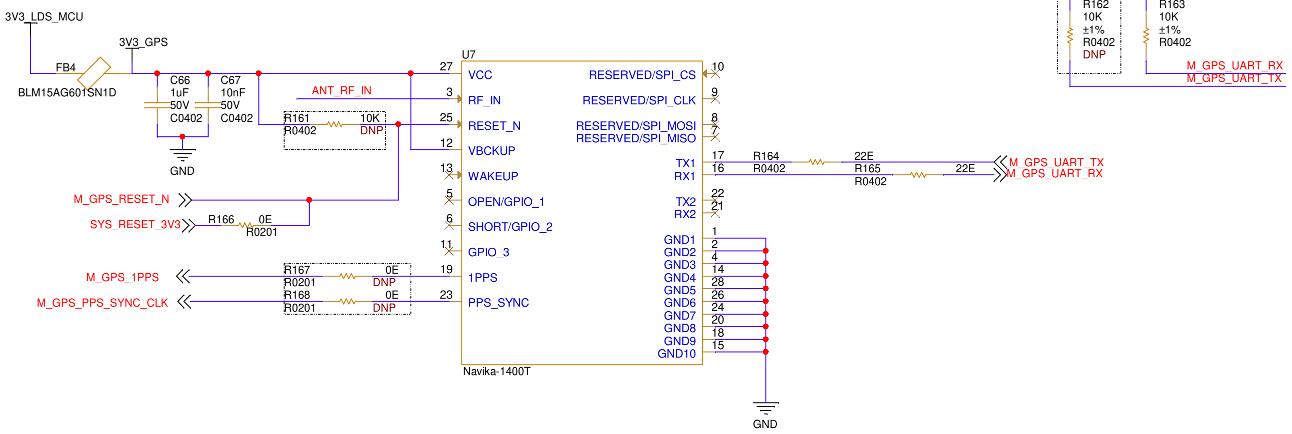
Author:	Arun P	Reviewed by:	<Reviewed By>

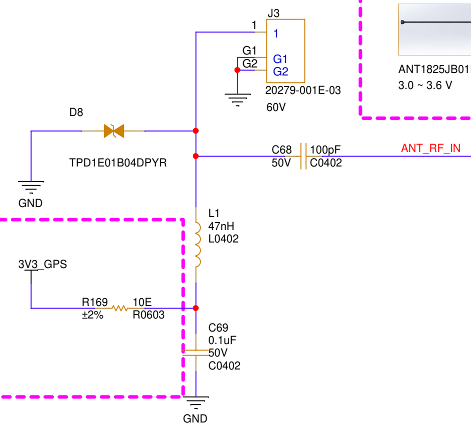
GPS/NAVIC INTERFACE

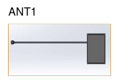
Design note:

1.Power pin De-caps has to be placed close to the pins
2.The "RF_IN" trace should match 50ohm impedence
3.Provide a 25us low pulse to reset the module



ANTENNA





NOTE:-
Length 25.0 ± 0.3 mm
Width 25.0 ± 0.3 mm
Thickness 12 mm (MAX.)
Cable Diameter = 1.13 mm ;
Length = 100 mm ± 2 mm ;
Black Connector I-PEX Connector 20278

NOTE:-
R116 --> 10 E resistor with 1.1W,
which reduces the voltage drop
to 0.15 V and provides a higher
antenna supply voltage of 3.15 V.

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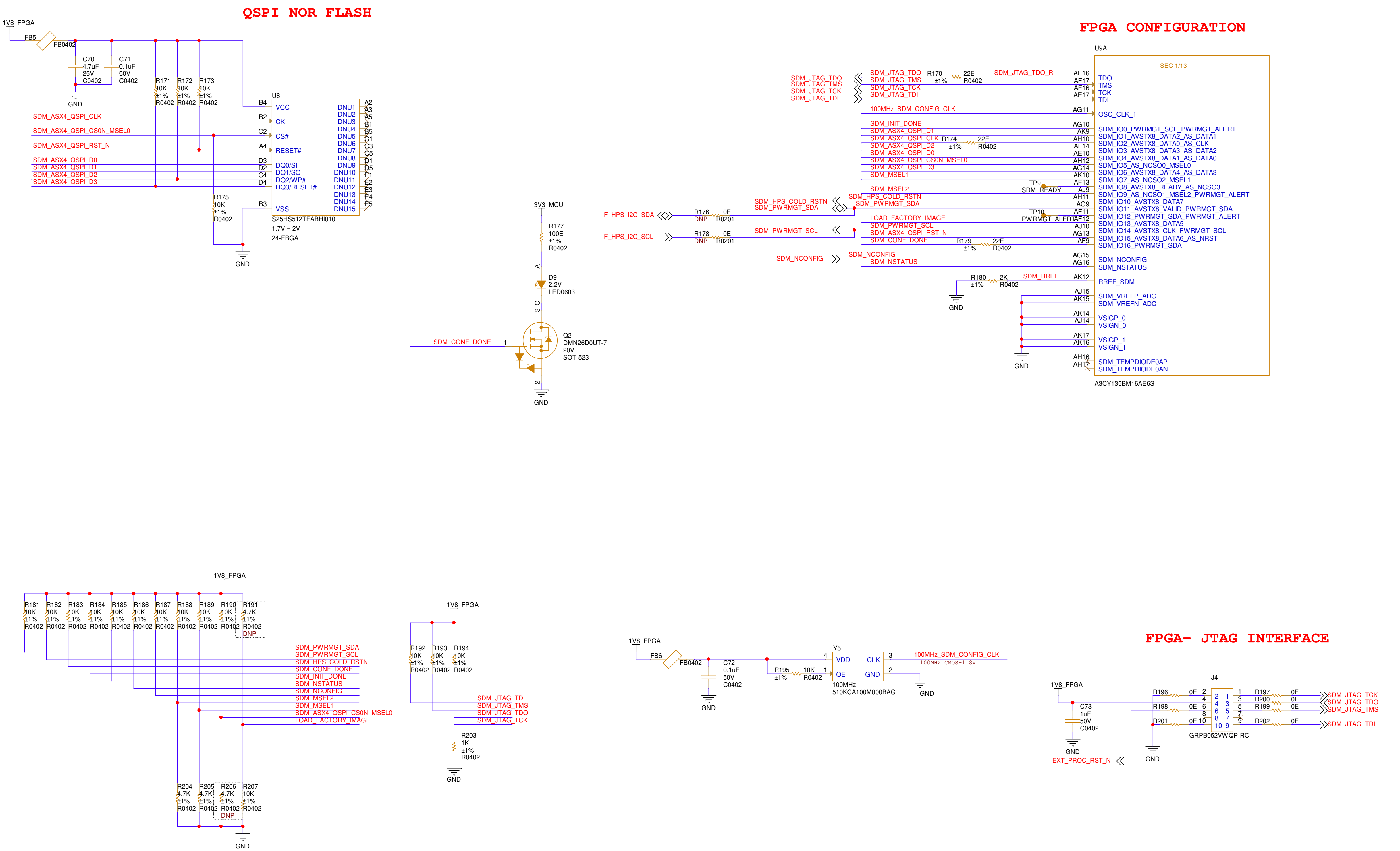
Title:	Handheld Radio Development Board
Sheet Name:	GPS/NAVIC Interface
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 11 of 31
Author: Arun P	Reviewed by: <Reviewed By>

FPGA-QSPI NOR FLASH INTERFACE



CONFIG MODE	MSEL2	MSEL1	MSEL0	
AS (NORMAL)	0	1	1	
AS (FAST)	0	0	1	<- DEFAULT

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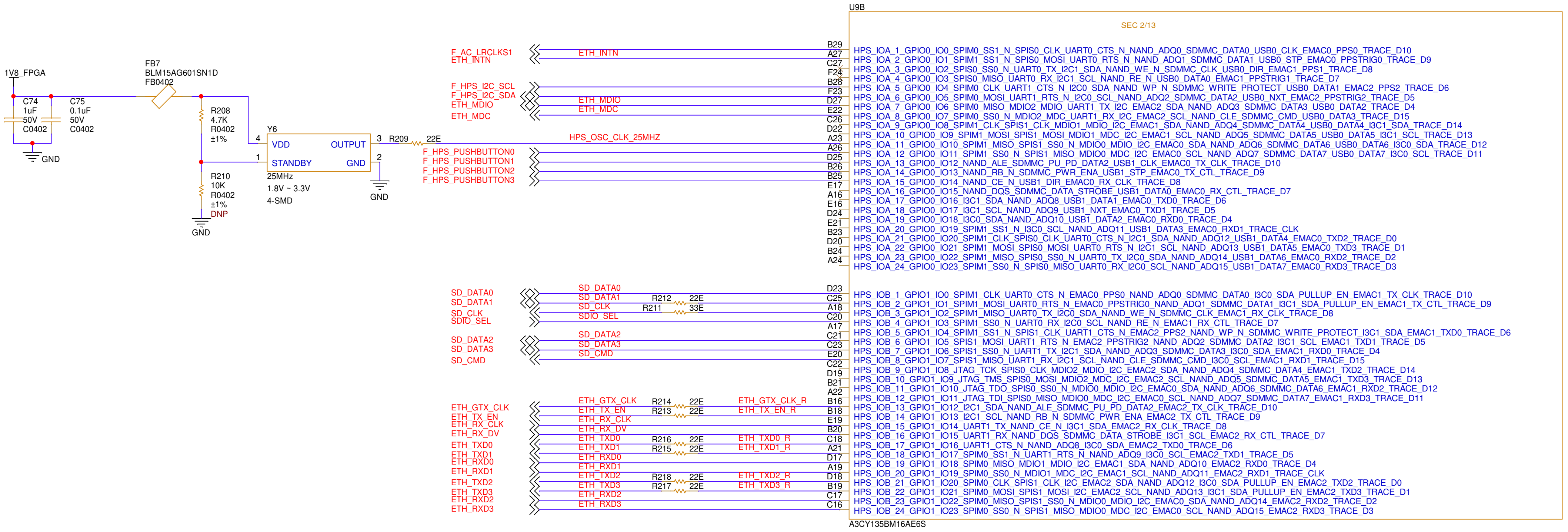
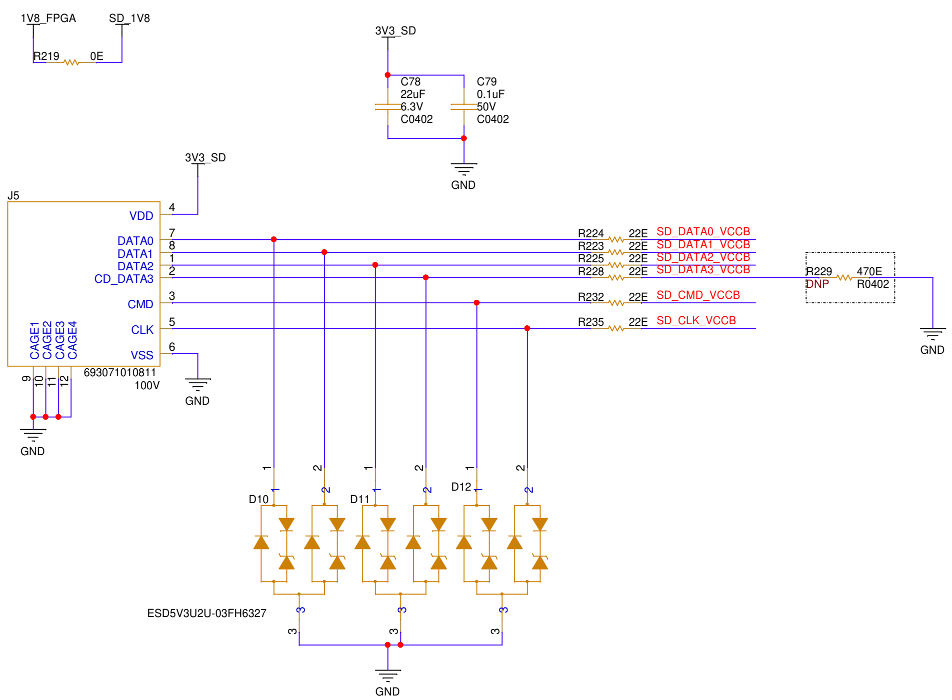
Title: Handheld Radio Deveploment Board
Sheet Name: FPGA-QSPI Nor Flash Interface
Size: C   REV: R0.1
Date Created: Friday, July 03, 2026   Sheet: 12 of 31
Author: Arun P   Reviewed by: <Reviewed By>

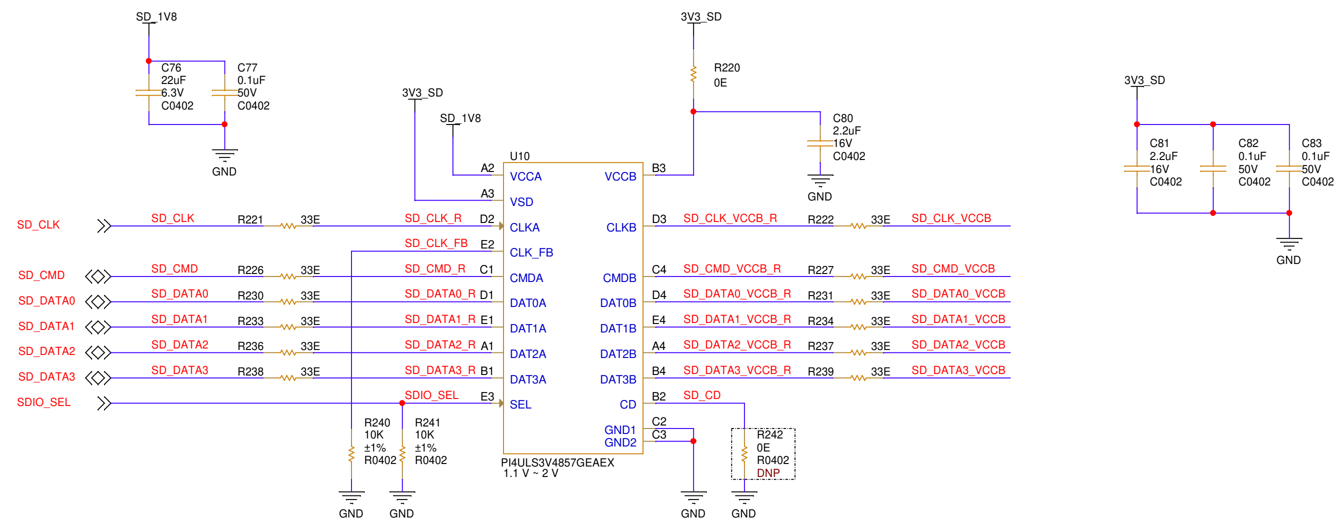
FPGA- MICRO SD CARD INTERFACE



MICRO SD CARD SOCKET



MICRO SD CARD LEVEL SHIFTER



SEL	VCCB
LOW (0V)	3.3V (DEFAULT)
HIGH (1.8V)	1.8V

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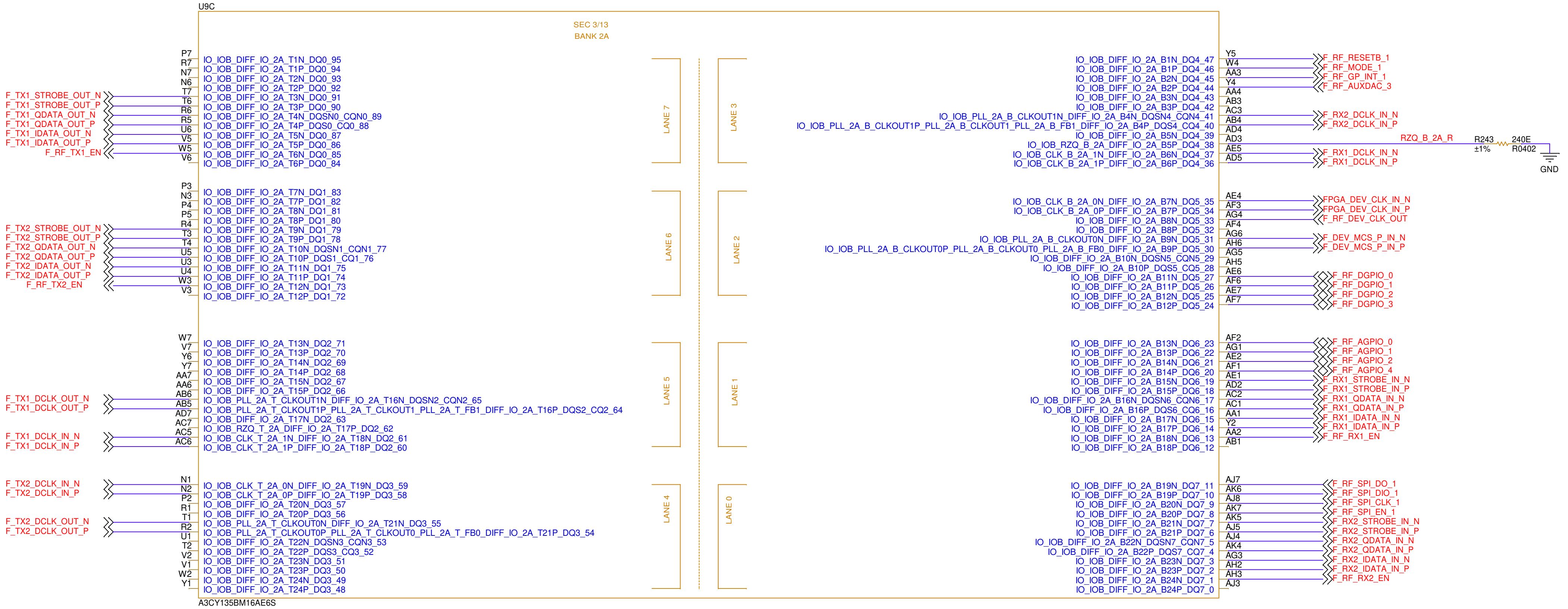
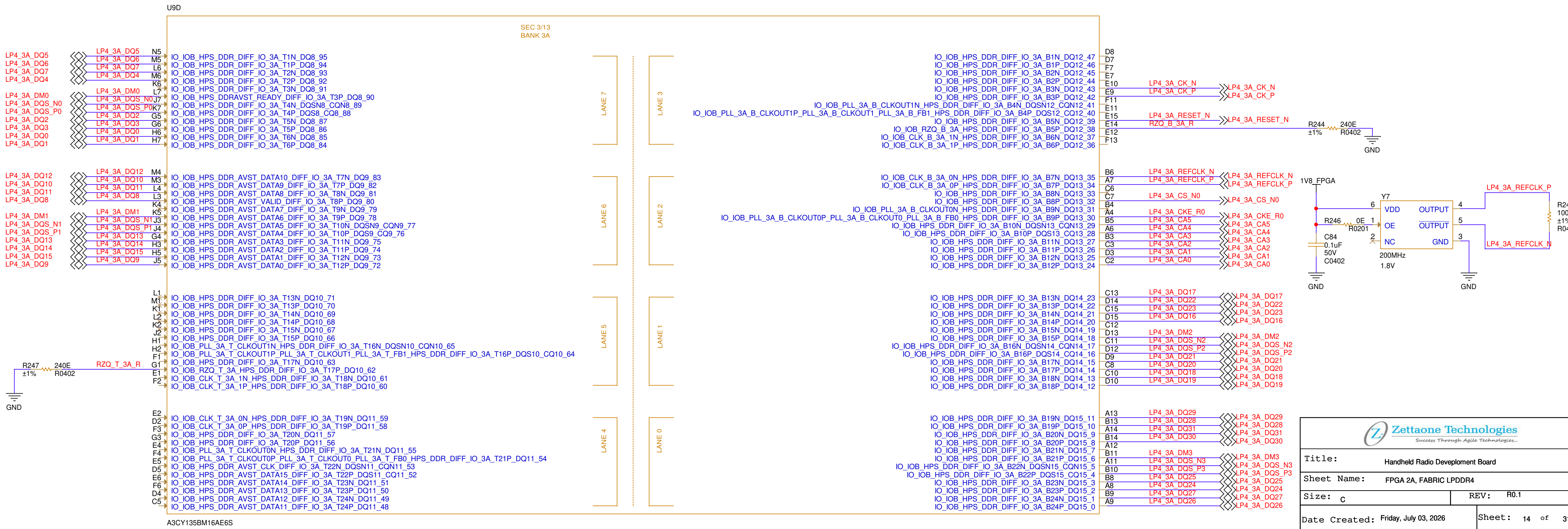
Title:	Handheld Radio Development Board
Sheet Name:	FPGA- Micro SD Card Interface
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 13 of 31
Author: Arun P	Reviewed by: <Reviewed By>

FPGA 2A, ADRV9002 RF TRANSCEIVER



FPGA 3A, HPS LPDDR4



Title: Handheld Radio Development Board

Sheet Name: FPGA 2A, FABRIC LPDDR4

Size: C | REV: R0.1

Date Created: Friday, July 03, 2026 | Sheet: 14 of 31

Author: Arun P | Reviewed by: <Reviewed By>

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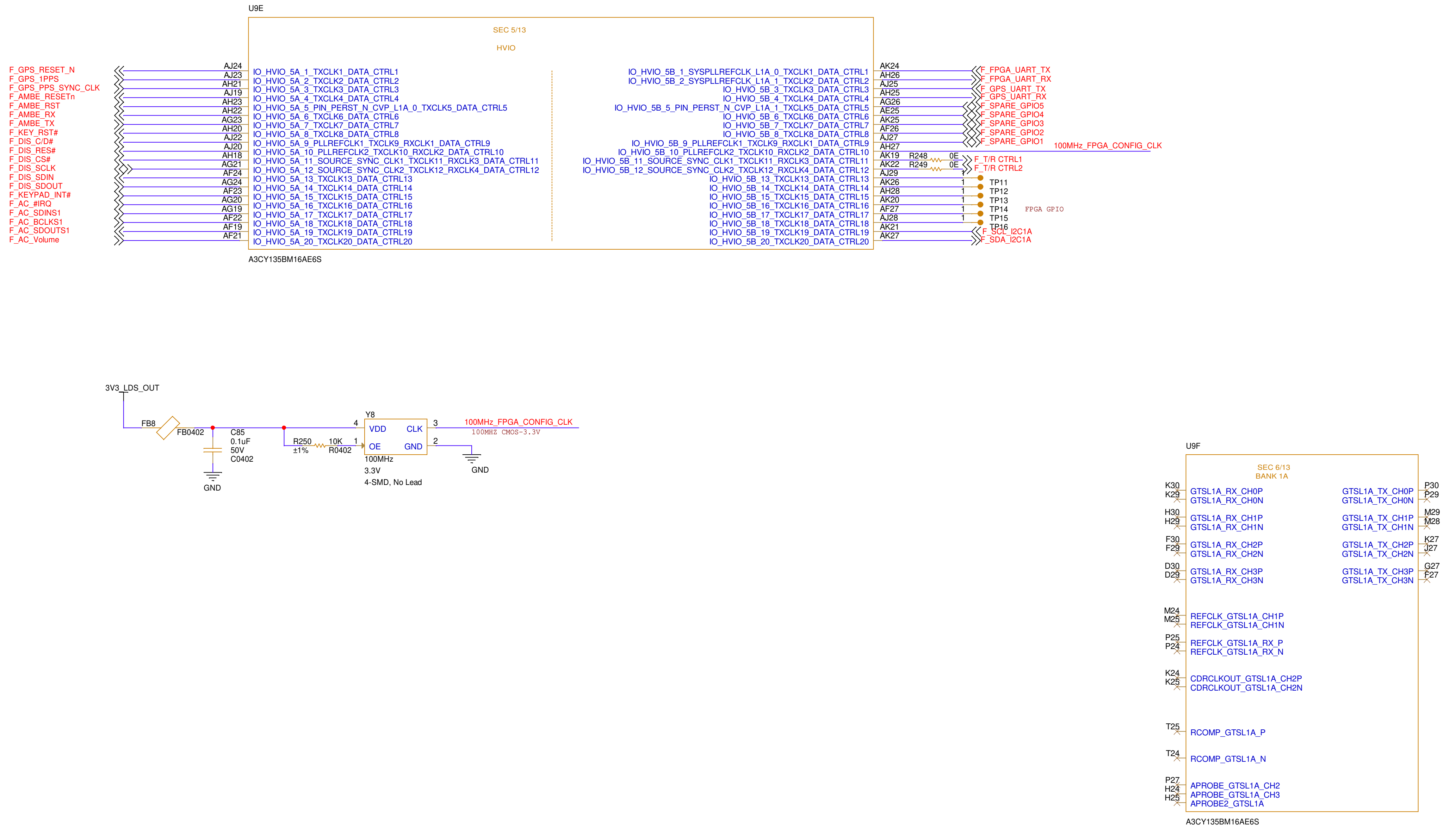
FPGA- INTERFACE



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Title:	Handheld Radio Development Board		
Sheet Name:	FPGA- Interface		
Size:	C	REV:	R0.1
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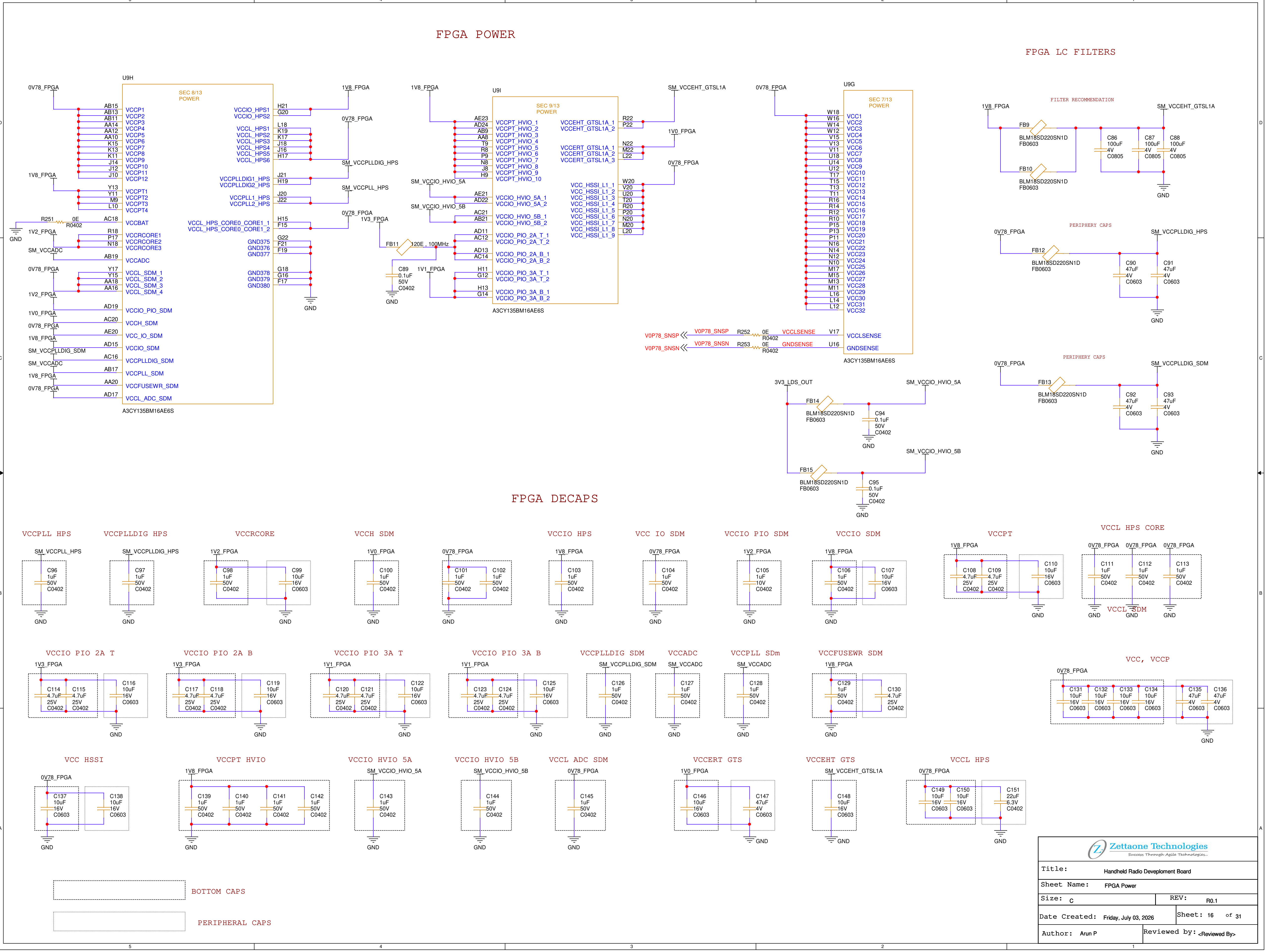

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Title:	Handheld Radio Devleploment Board
Sheet Name:	FPGA Power
Size: C	REV: R0.1
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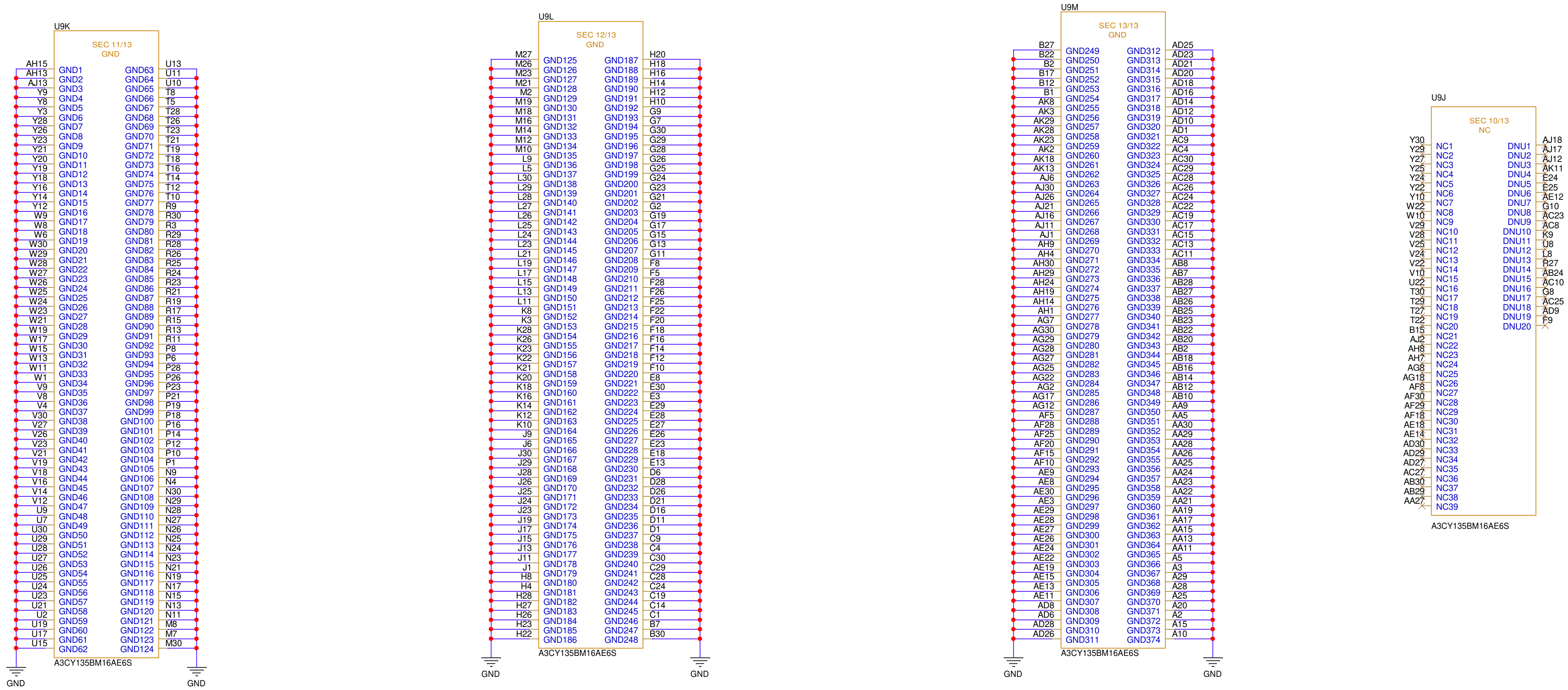
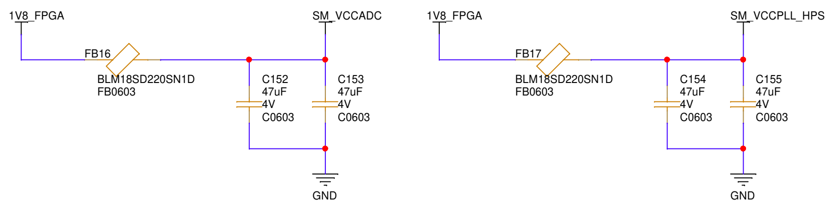
Author: Arun P	Reviewed by: <Reviewed By>

FPGA GND



FPGA LC FILTERS



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Title: Handheld Radio Develpoment Board
Sheet Name: FPGA GND
Size: C
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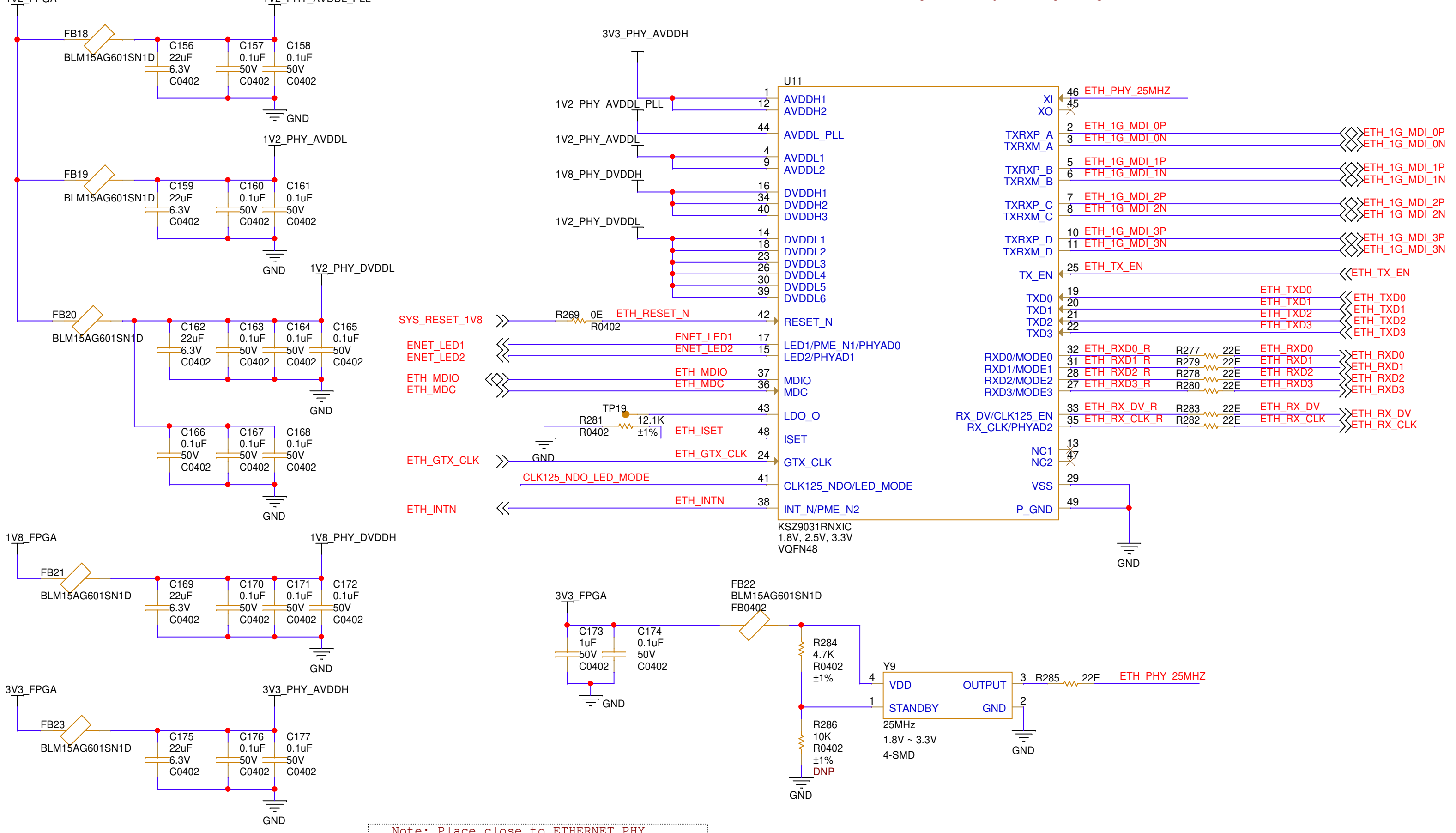
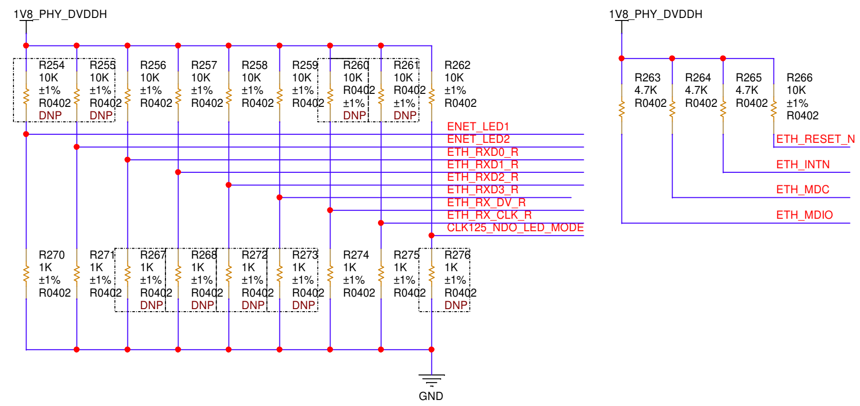
Author: Arun P
Reviewed by: <Reviewed By>

ETHERNET PHY-POWER & DECAPS

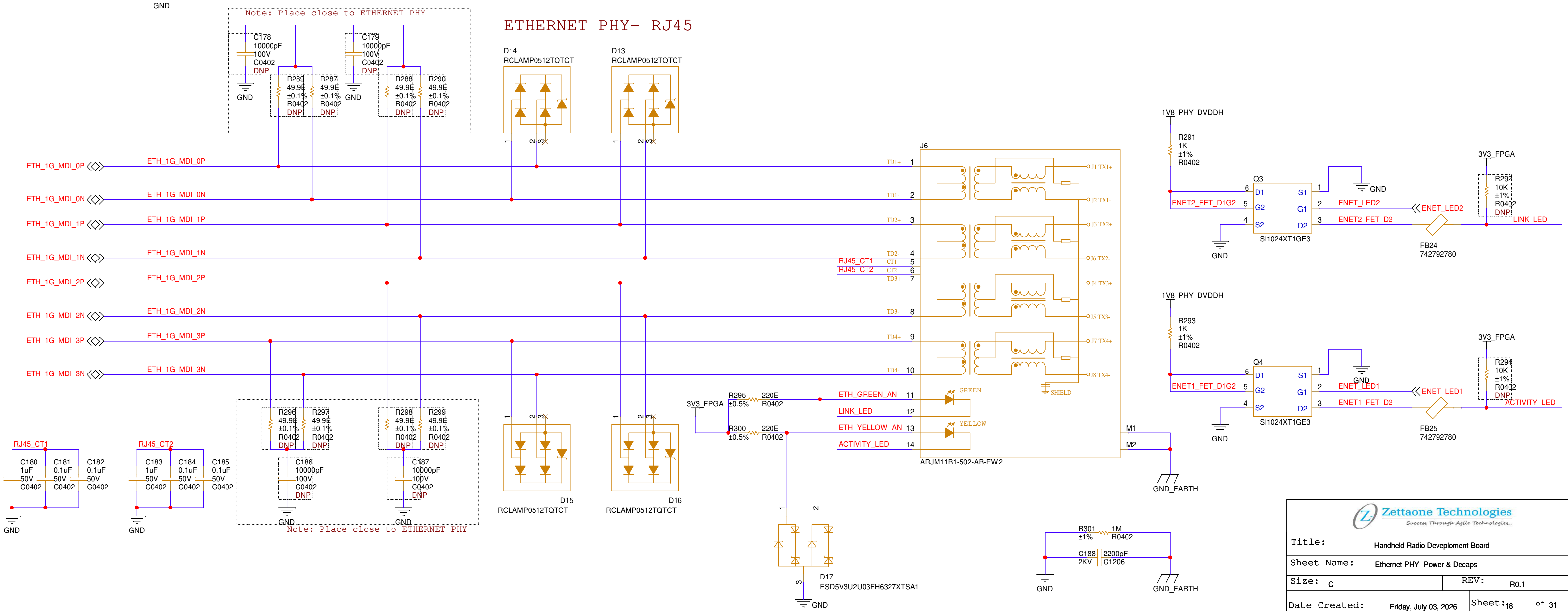




ETHERNET INTERFACE

NAME	PIN	VALUE	DESCRIPTION
PHYAD2	35	0	ID = 0B00000
PHYAD1	15	0	"RGMII MODE - ADVERTISE ALL CAPABILITIES (10/100/1000 SPEED HALF-/FULL-DUPLEX), EXCEPT 1000BASE-T HALF DUPLEX"
PHYAD0	17	0	
MODE3	27	1	
MODE2	28	1	
MODE1	31	1	
MODE0	32	1	
CLK125_EN	33	0	NO CLOCKOUTPUT FROM PIN41
LED_MODE	41	1	SINGLE LED MODE

ETHERNET PHY- RJ45



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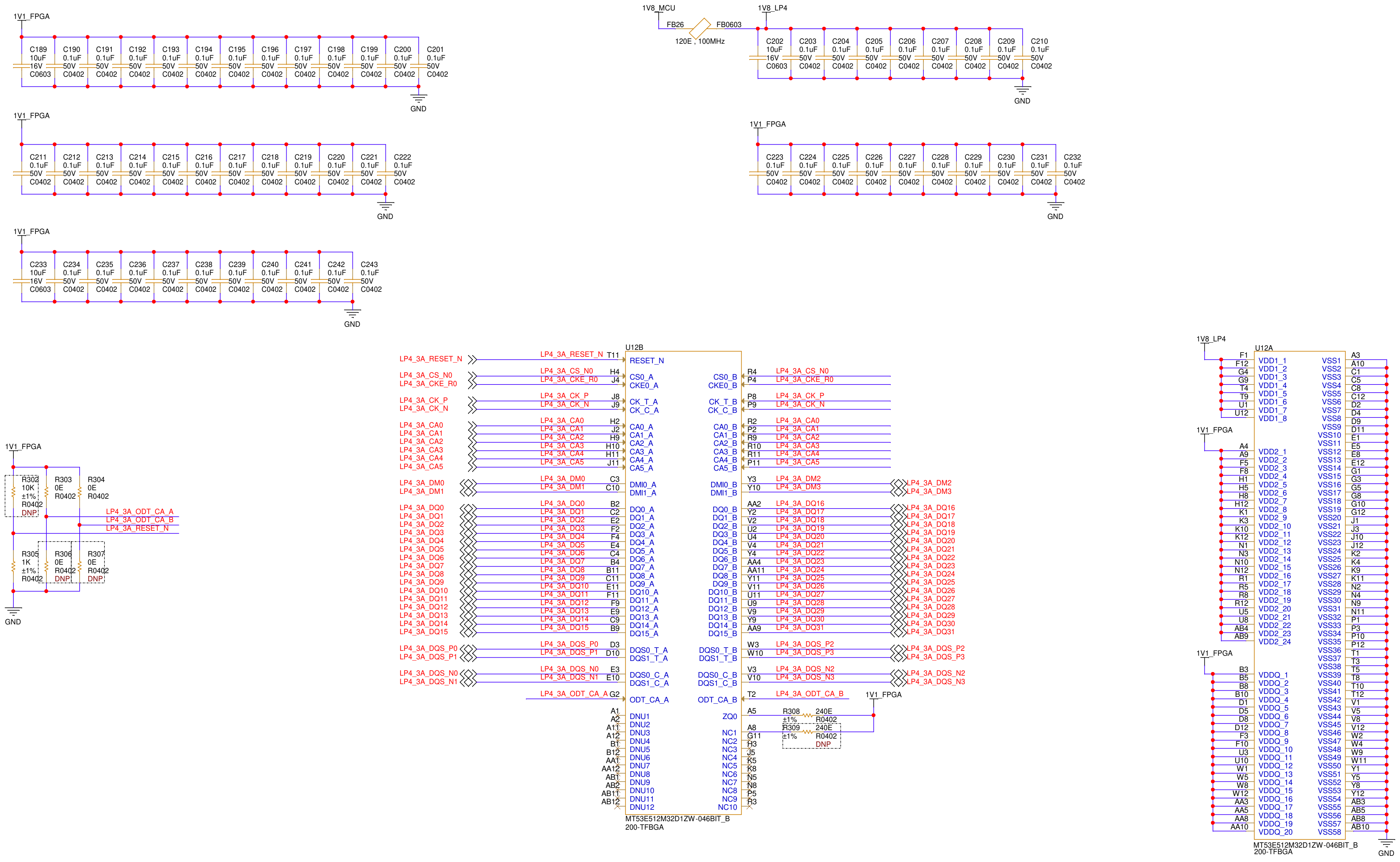
Title:	Handheld Radio Deveploment Board
Sheet Name:	Ethernet PHY- Power & Decaps
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 18 of 31
Author: Arun P	Reviewed by: <Reviewed By>

HPS LPDDR4 INTERFACE



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Title:	Handheld Radio Deveplopment Board
Sheet Name:	HPS LPDDR4 Interface
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 19 of 31
Author: Arun P	Reviewed by: <Reviewed By>

RF-FPGA INTERFACE

IMPEDANCE CHARACTERISTICS:
ALL DATA PORT DIFFERENTIAL PAIRS = 100ohm DIFFERENTIAL,

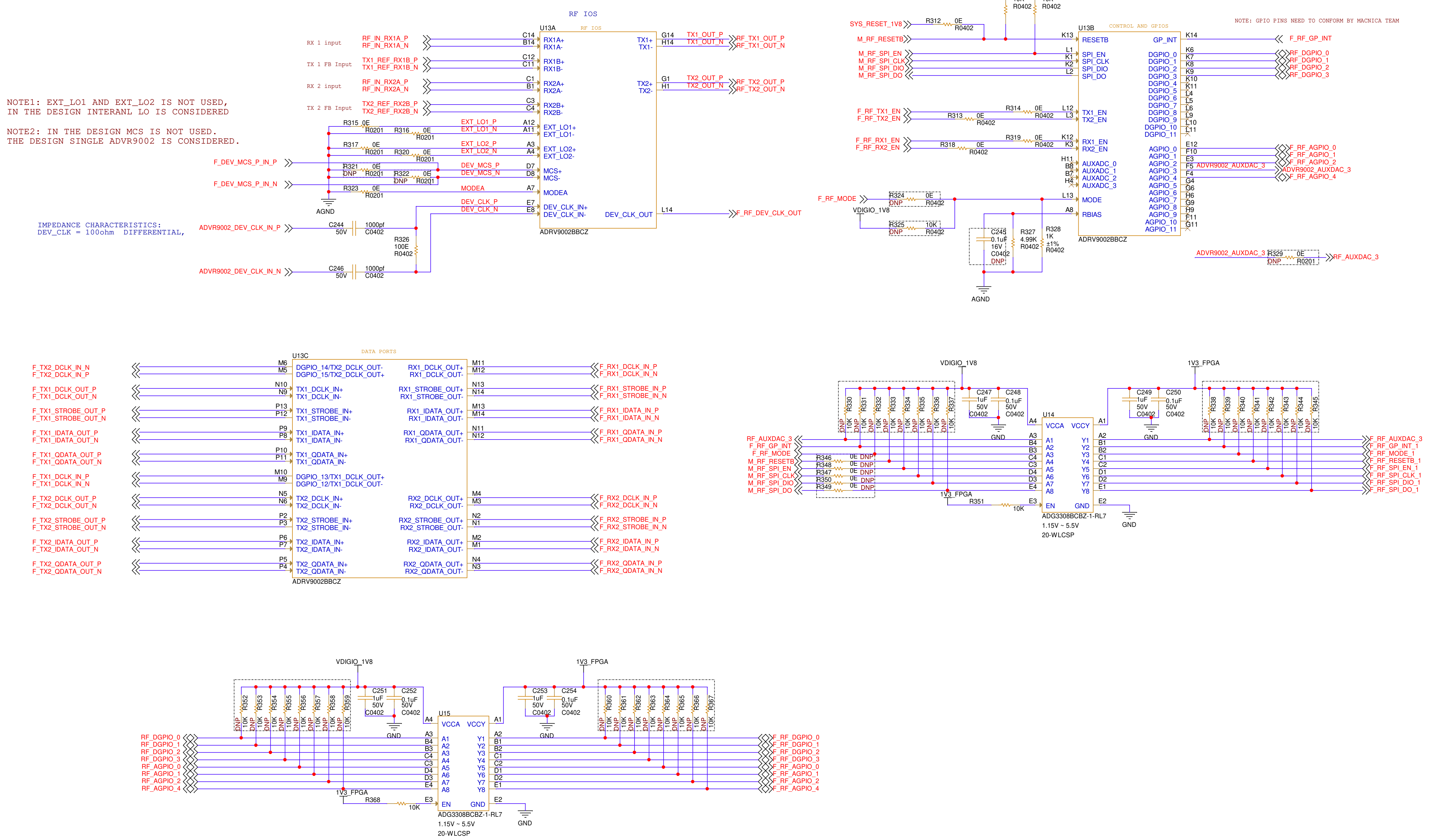
RF-CONTROL AND GPIOS

NOTE: GPIO PINS NEED TO CONFORM BY MACNICA TEAM

NOTE1: EXT_LO1 AND EXT_LO2 IS NOT USED,
IN THE DESIGN INTERANL LO IS CONSIDERED

NOTE2: IN THE DESIGN MCS IS NOT USED.
THE DESIGN SINGLE ADRV9002 IS CONSIDERED.

IMPEDANCE CHARACTERISTICS:
DEV_CLK = 100ohm DIFFERENTIAL,



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Title: Handheld Radio Development Board
Sheet Name: RF-FPGA Interface
Size: C
REV: R0.1
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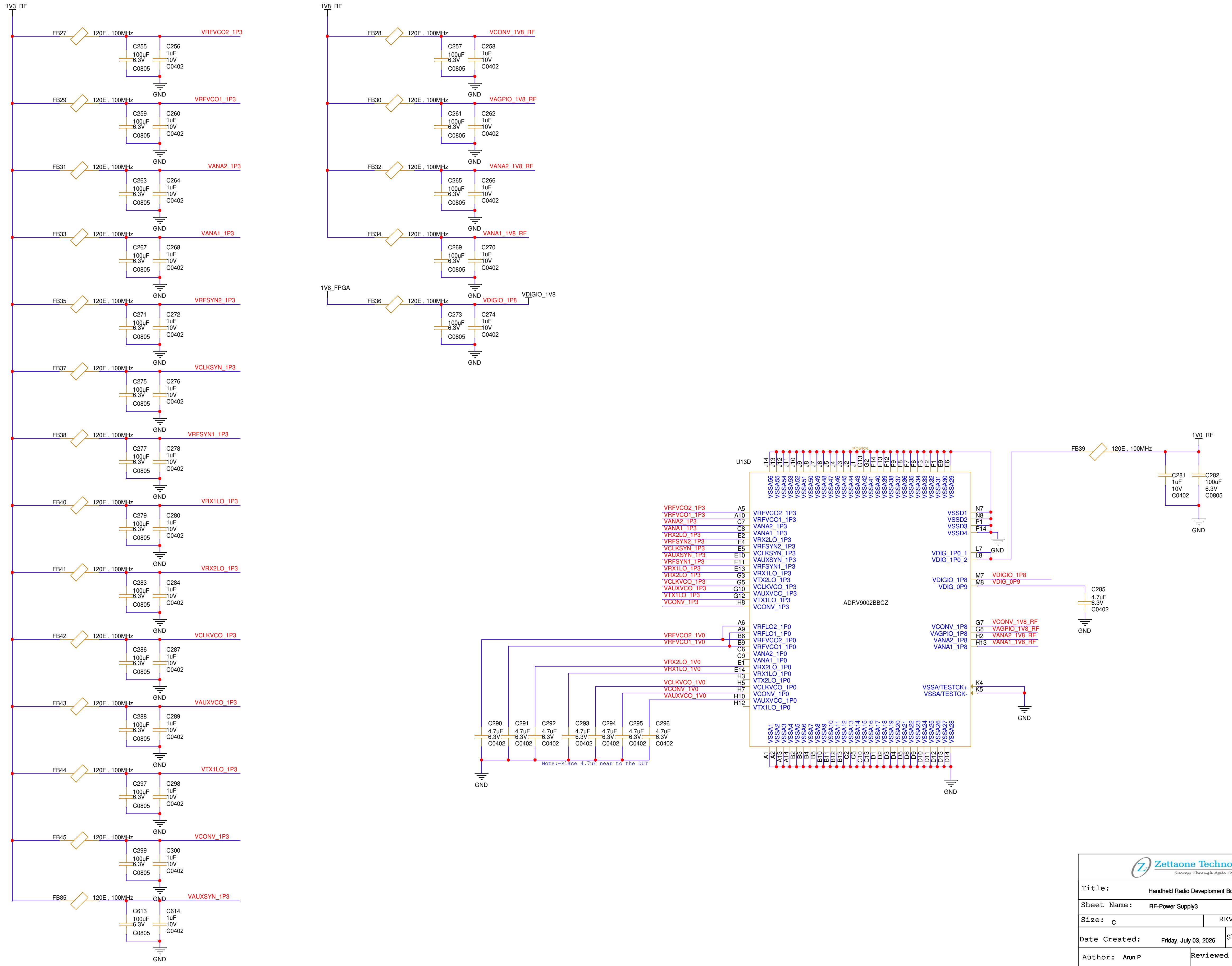
Author: Arun P
Reviewed by: <Reviewed By>

RF-POWER DECAPS



Title:	Handheld Radio Development Board
Sheet Name:	RF-Power Supply3
Size: C	REV: R0.1
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Author: Arun P	Reviewed by: <Reviewed By>

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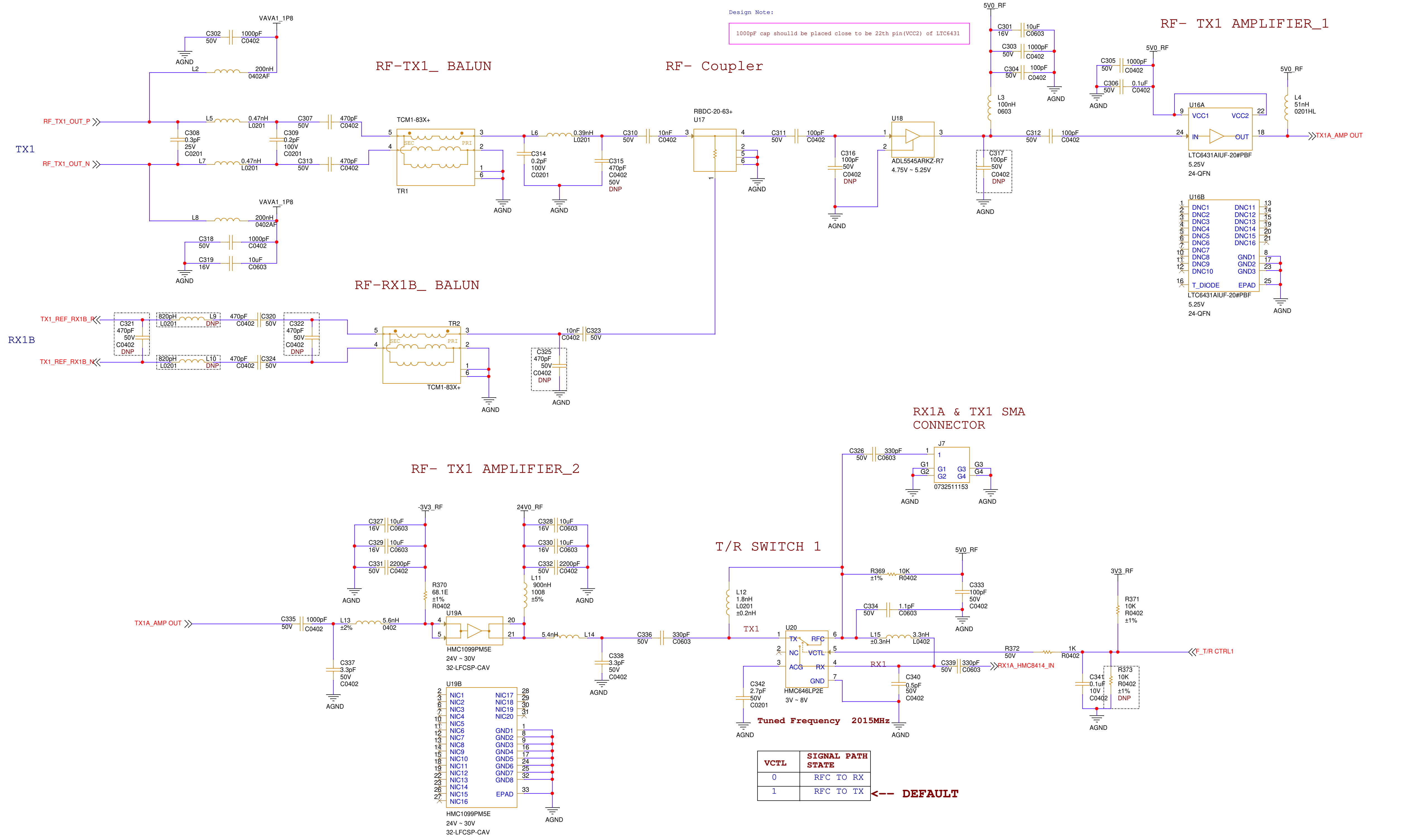

VCTL	SIGNAL PATH STATE
0	RFC TO RX
1	RFC TO TX

<-- DEFAULT

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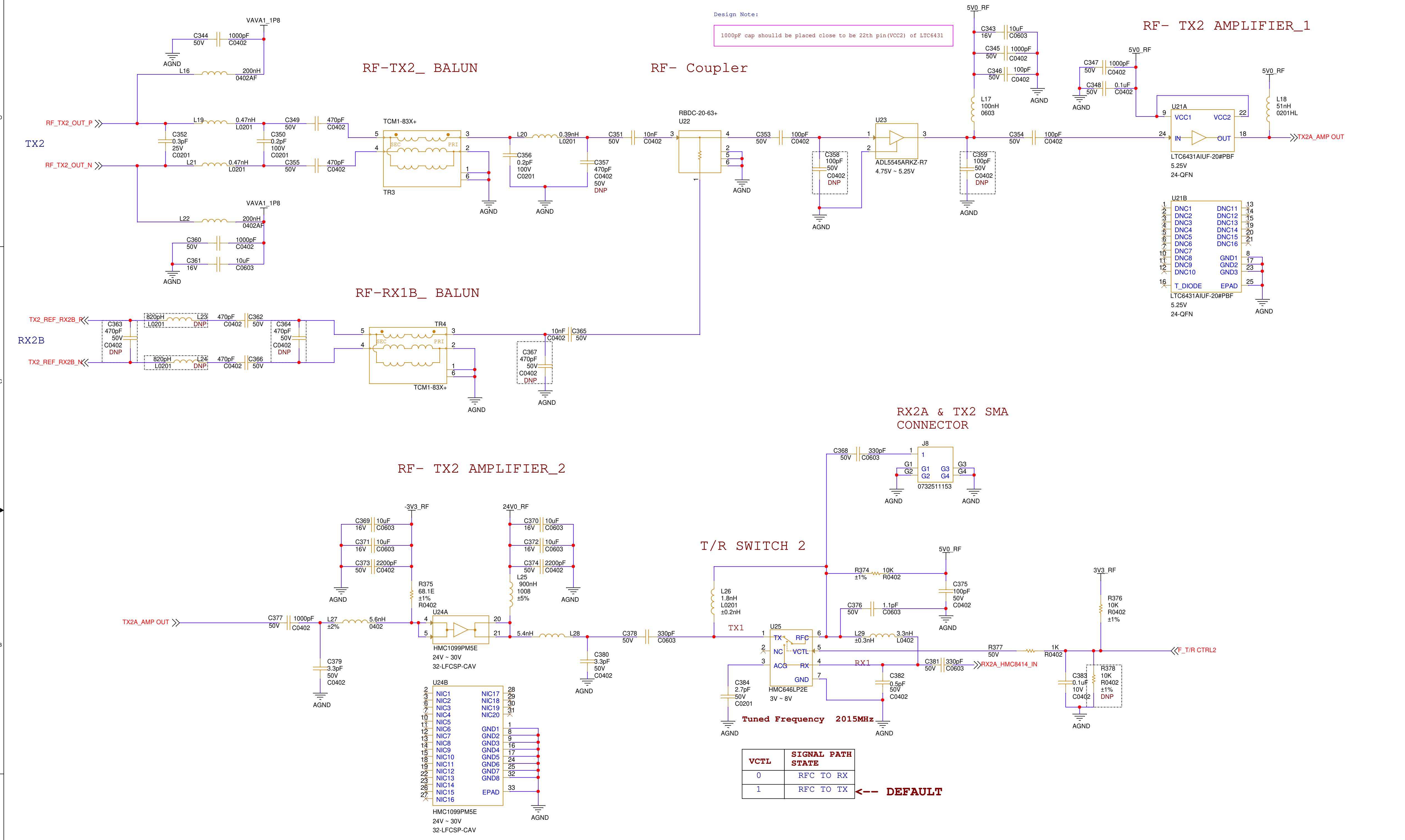
Title:	Handheld Radio Development Board
Sheet Name:	RF-BALUN_1
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 22 of 31
Author: Arun P	Reviewed by: <Reviewed By>



VCTL	SIGNAL PATH STATE
0	RFC TO RX
1	RFC TO TX

<-- DEFAULT

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Title: Handheld Radio Development Board

Sheet Name: RF-TX Amplifier_1

Size: C

REV: R0.1

Date Created: Friday, July 03, 2026

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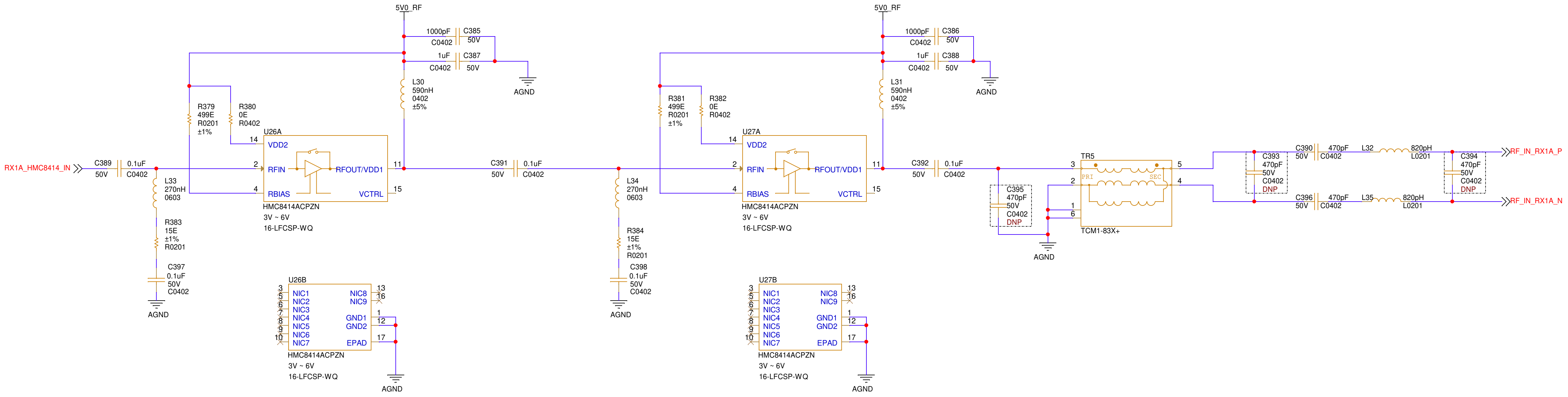
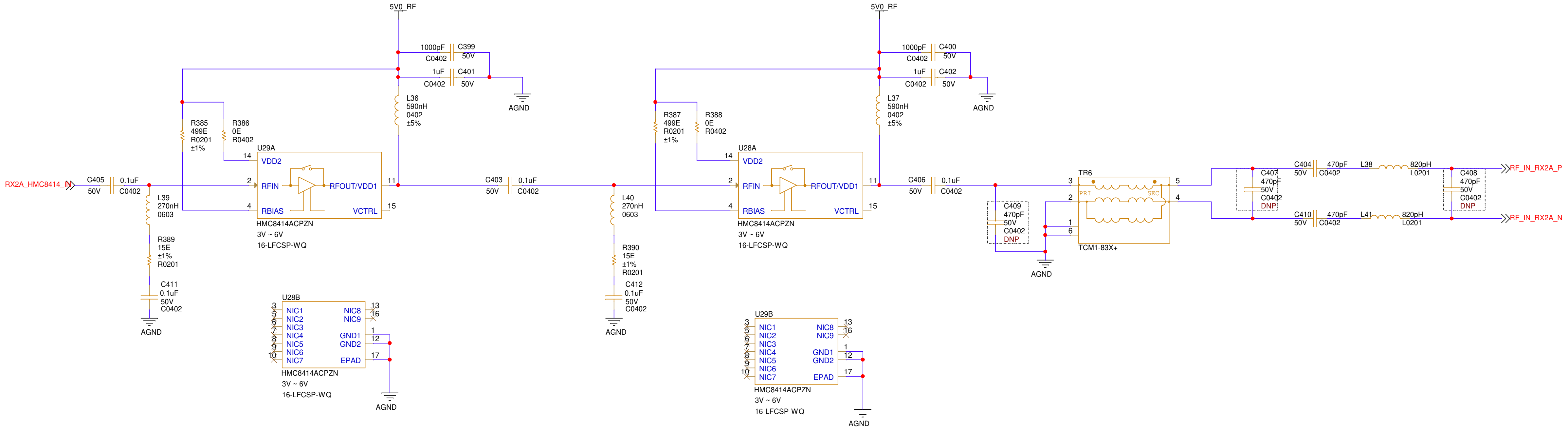
Author: Arun P

Reviewed by: <Reviewed By>

RX1A LNA & BALUN SECTION



RX1A LNA & BALUN SECTION



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Title:	Handheld Radio Development Board
Sheet Name:	RF-RX Amplifier_1
Size: C	REV: R0.1
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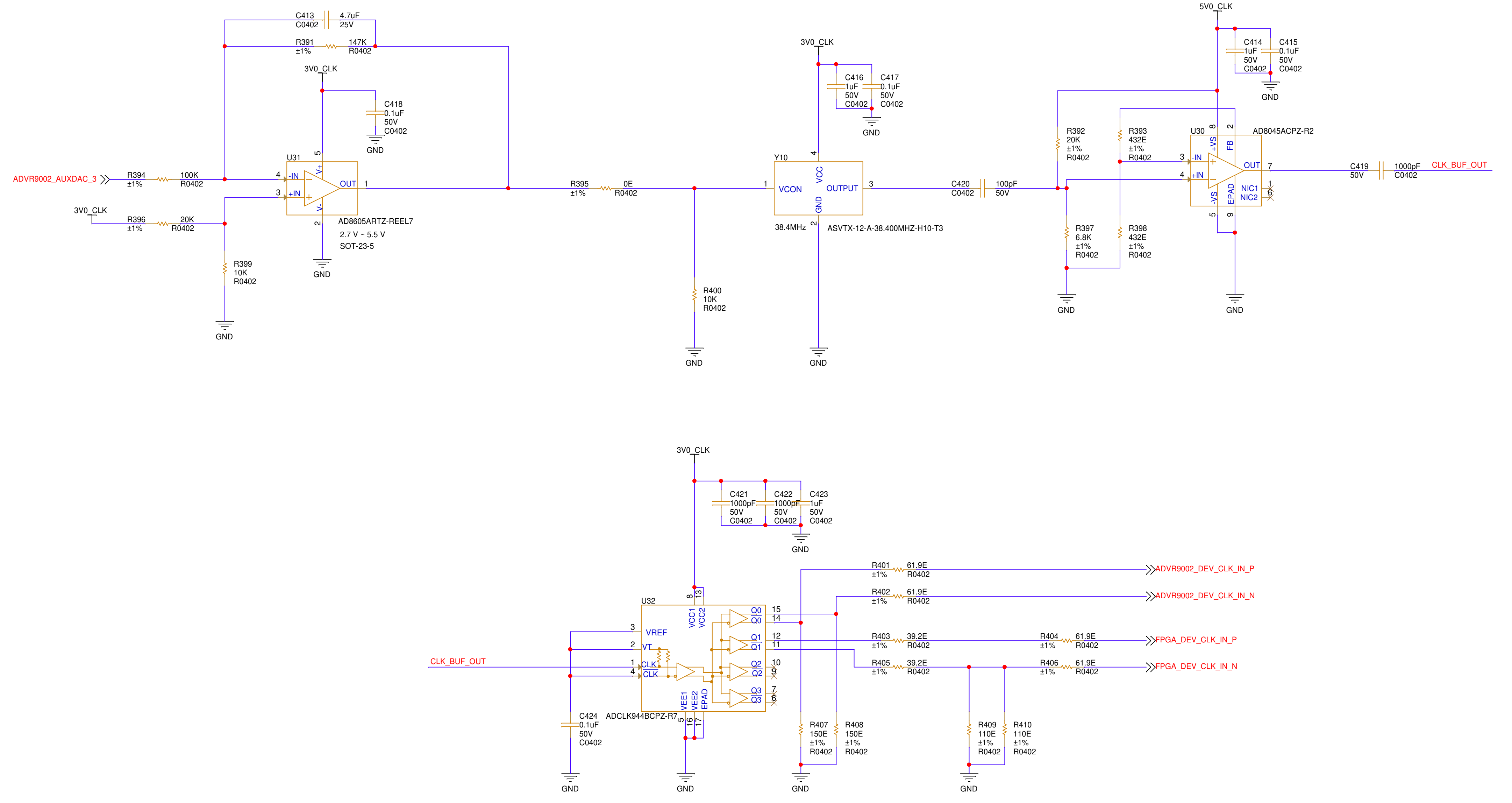
Author: Arun P	Reviewed by: <Reviewed By>

RF-CLOCK DISTRIBUTION



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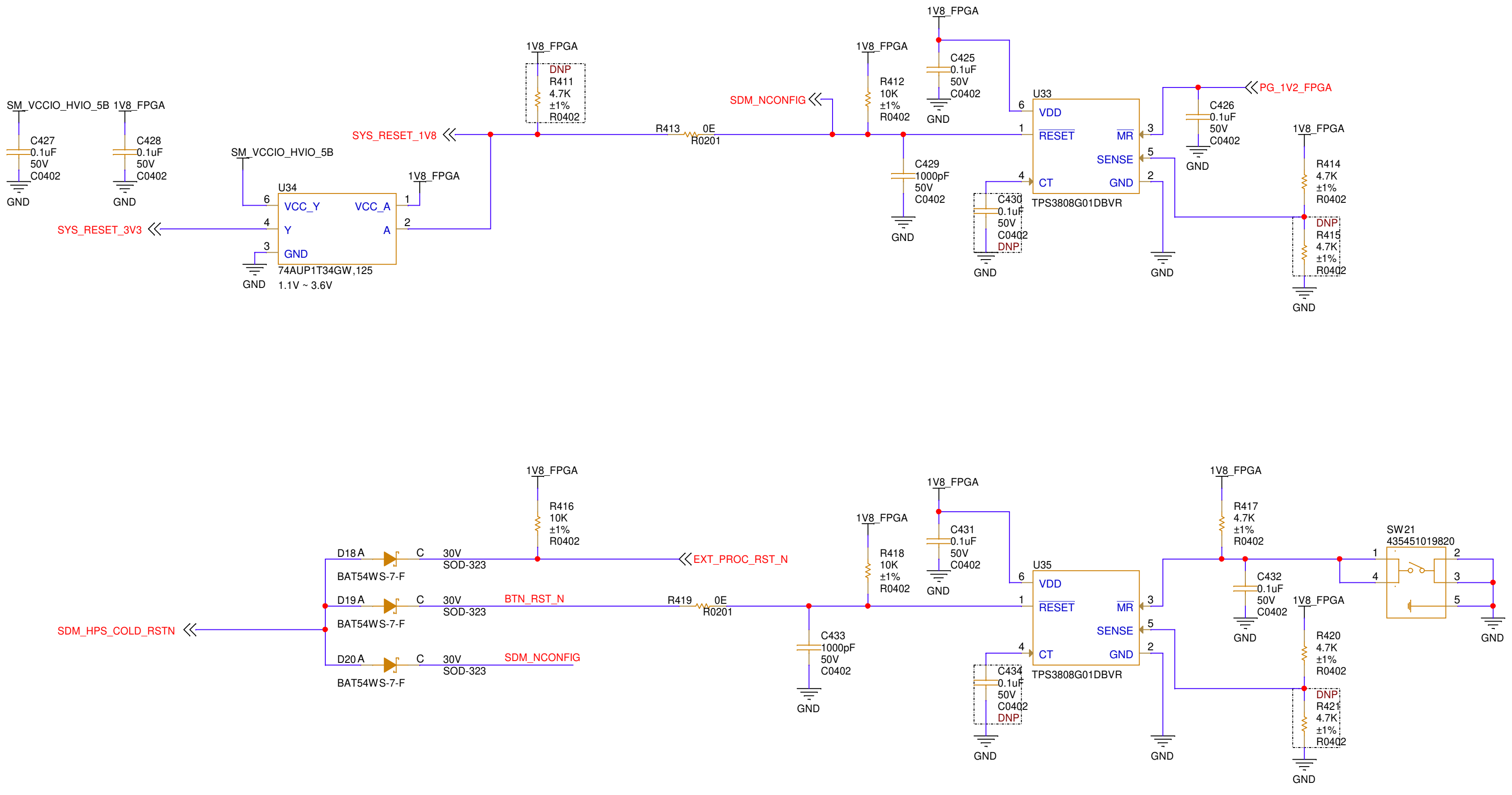
Title:	Handheld Radio Development Board
Sheet Name:	RF-Power Supply2
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 25 of 31
Author: Arun P	Reviewed by: <Reviewed By>

SYSTEM RESET FOR 1V8 AND 3V3



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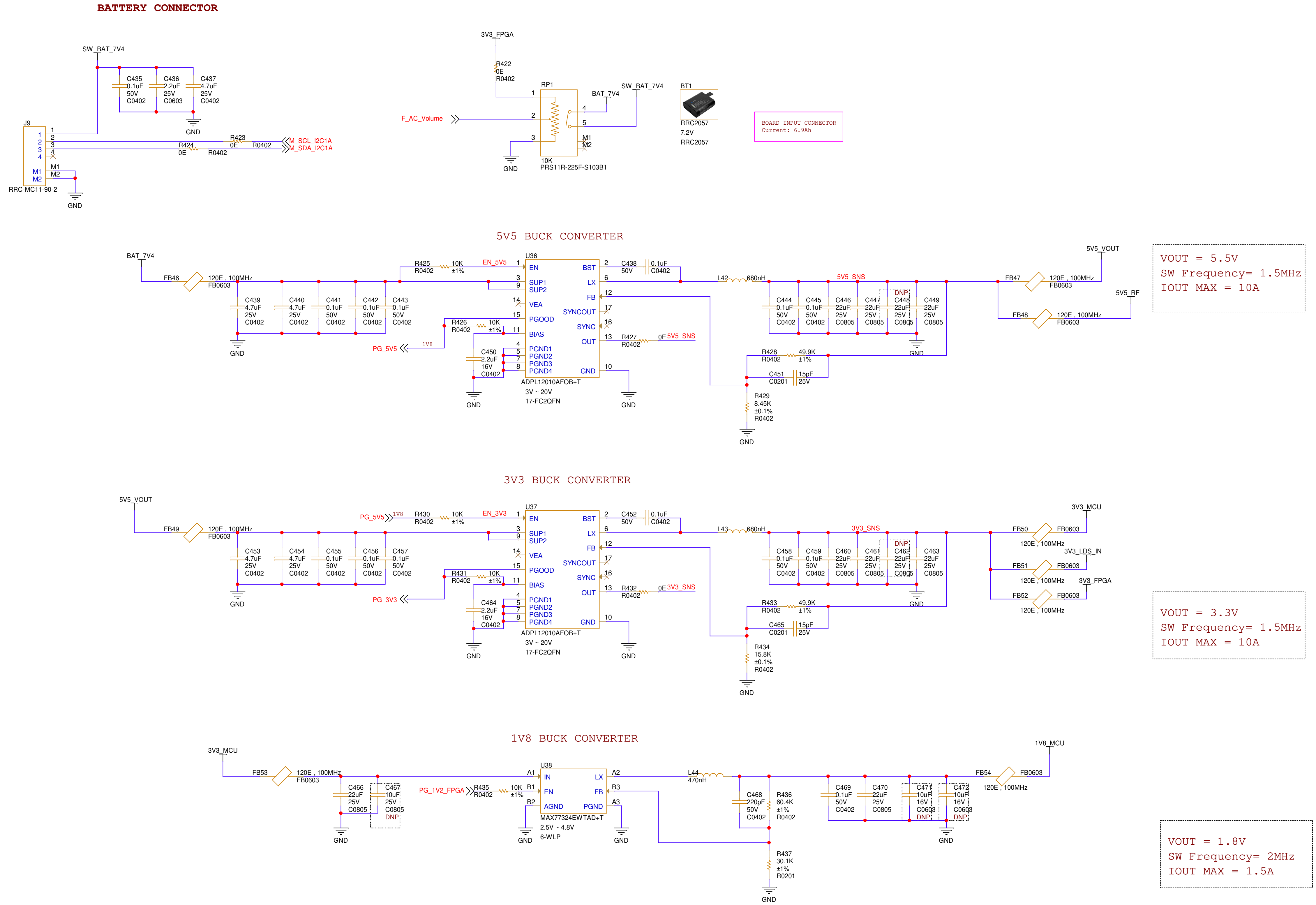
Title:	Handheld Radio Development Board
Sheet Name:	RF-Power Decaps
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 26 of 31
Author: Arun P	Reviewed by: <Reviewed By>



Zettaone Technologies
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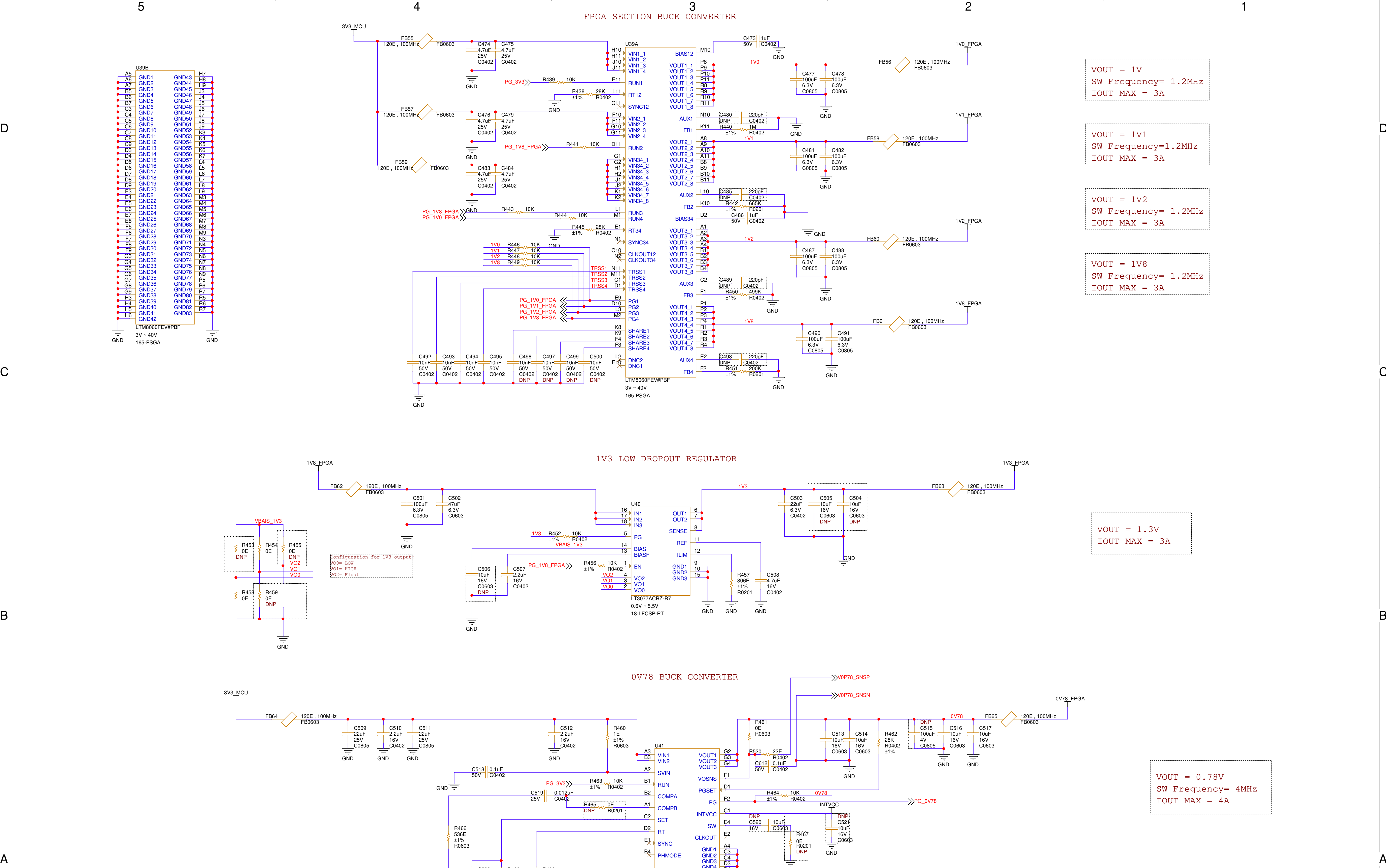
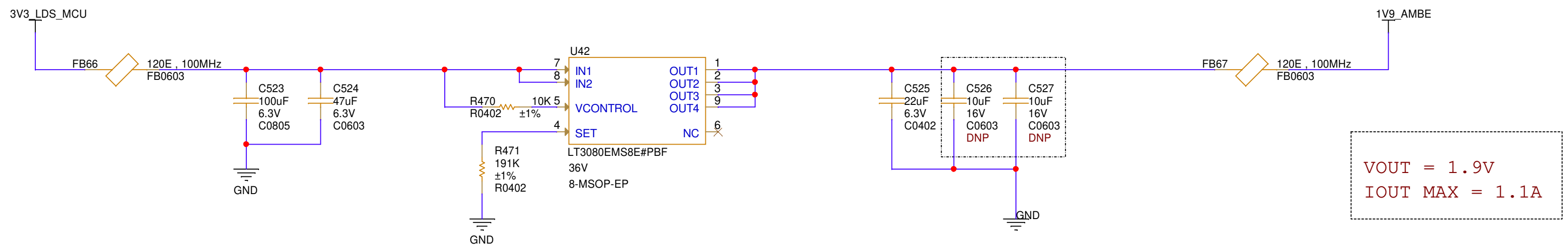
Title:	Handheld Radio Development Board		
Sheet Name:	MCU Power supply		
Size:	C	REV:	R0.1
Date Created:	Friday, July 03, 2026	Sheet:	27 of 31
Author:	Arun P	Reviewed by:	<Reviewed By>

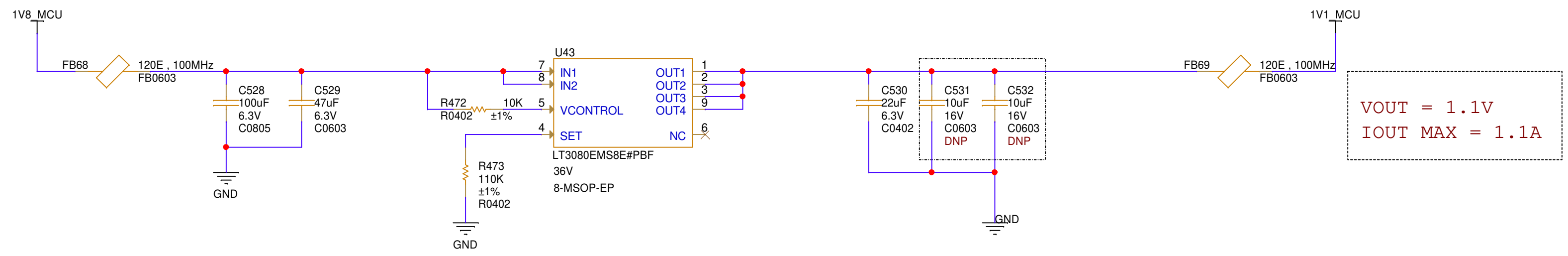


Zettaone Technologies — Success Through Agile Technologies	
Title:	Handheld Radio Devoplement Board
Sheet Name:	FPGA Power supply
Size: A2	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 28 of 31
Author: Arun P	Reviewed by: <Reviewed By>

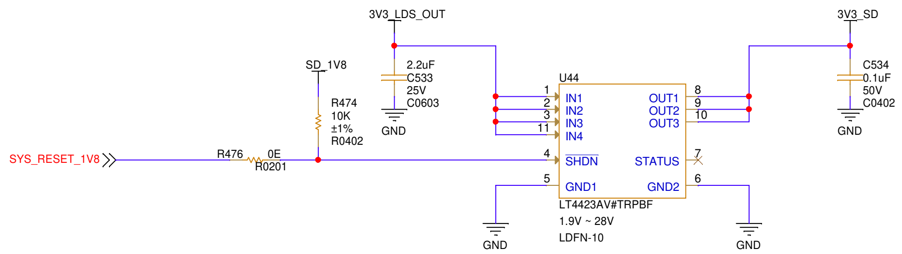
1V9 LOW DROPOUT REGULATOR



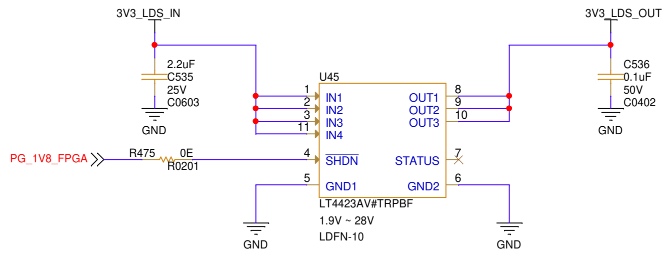
1V1 LOW DROPOUT REGULATOR



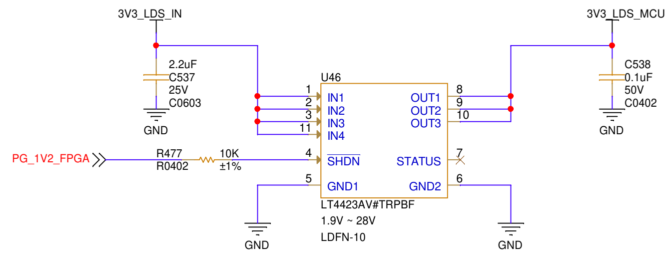
SD CARD LOAD SWITCH



3V3 FPGA 5A/5B BANK LOAD SWITCH



3V3 CONTROLER LOAD SWITCH



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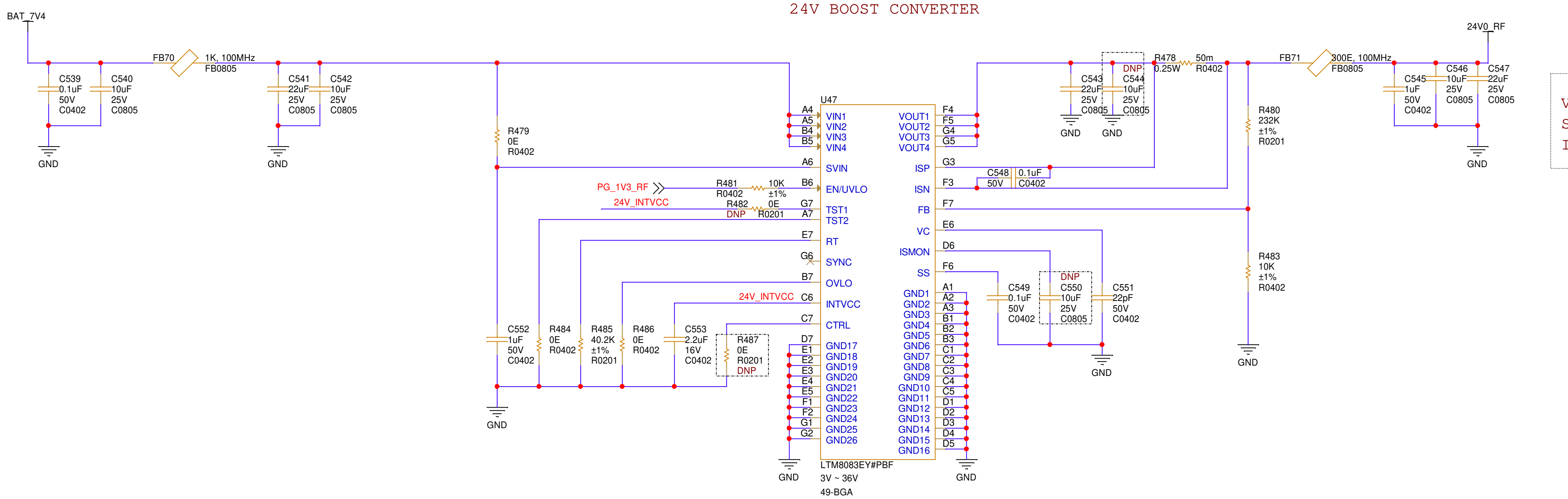
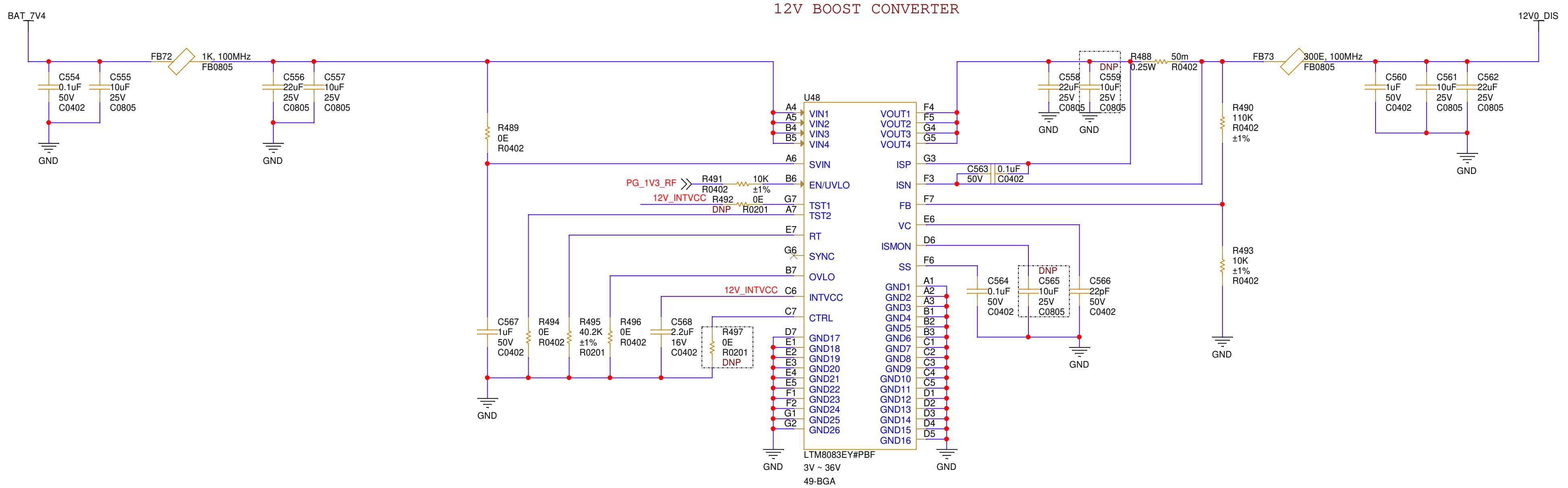
Title:	Handheld Radio Devleploment Board
Sheet Name:	MCU & Display Power supply
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 29 of 31
Author: Arun P	Reviewed by: <Reviewed By>

24V BOOST CONVERTER



VOUT = 24V
SW Frequency= 1MHz
IOUT MAX = 1.5A

12V BOOST CONVERTER



VOUT = 12V
SW Frequency= 1MHz
IOUT MAX = 1.5A

Zettaone Technologies
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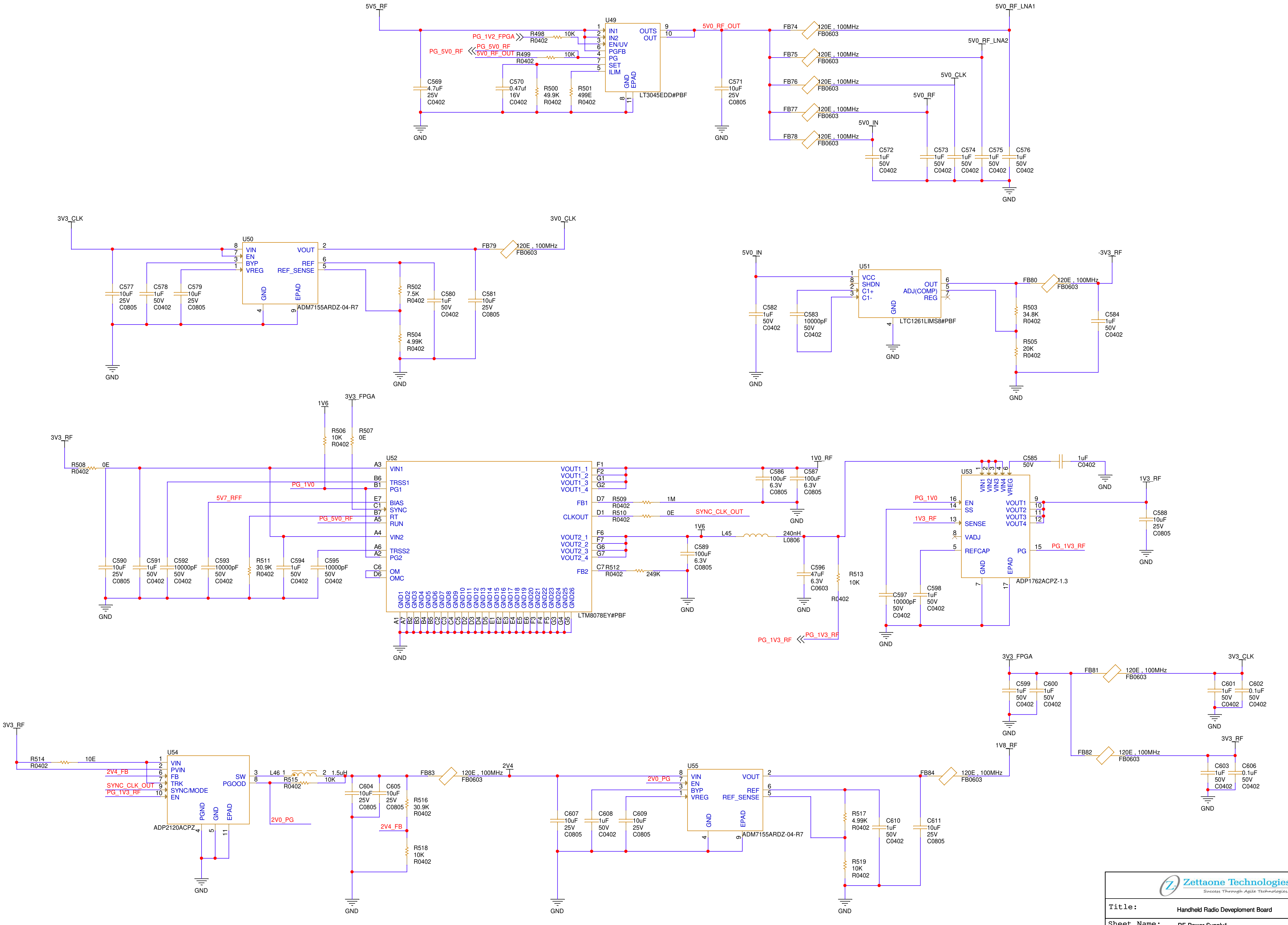
Title:	Handheld Radio Development Board
Sheet Name:	RF Power supply
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 30 of 31
Author: Arun P	Reviewed by: <Reviewed By>

RF-POWER SUPPLY1



Zettaone Technologies — Success Through Agile Technologies...	
Title:	Handheld Radio Development Board
Sheet Name:	RF-Power Supply1
Size: C	REV: R0.1
Date Created: Friday, July 03, 2026	Sheet: 31 of 31
Author: Arun P	Reviewed by: <Reviewed By>